

CI/CD Deployment Assignment

By: Muhammed Sahal SR

MSc Computer Science | AI/ML & DevOps Enthusiast

Project Title: Flask + Express CI/CD Deployment on AWS using Jenkins

Date: October 2025

Overview

This project demonstrates the deployment and automation of a Flask backend and an Express frontend application on an Amazon EC2 instance, along with a CI/CD pipeline implemented using Jenkins.

The objective was to automate deployment through Jenkins pipelines and ensure that every new code commit is deployed automatically and efficiently.

PART 1: Deploy Flask and Express on a Single EC2 Instance

Objective

Deploy both the Flask backend and the Express frontend on a single Amazon EC2 instance and make them accessible through public IP addresses.

1. Provisioning the EC2 Instance

Steps:

Logged into the AWS Management Console.

Launched a free-tier Ubuntu 22.04 LTS EC2 instance.

Configured security groups to allow:

Port 22 (SSH)

Port 5000 (Flask)

Port 3000 (Express)

Port 8080 (Jenkins)

SSH into the instance using:

```
ssh -i "mohdsahal924.pem" ubuntu@<EC2-Public-IP>
```

2. Installing Dependencies

Installed all required tools and runtimes on EC2:

```
sudo apt update -y
sudo apt install python3 python3-pip git -y
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -
sudo apt install -y nodejs
sudo npm install -g pm2
```

Installed Versions:

Python: 3.12.3

Node.js: 20.x

npm: 10.x

PM2: 5.x

3. Application Setup

✅ Cloned Project Repository:

```
git clone https://github.com/0079567603568sahal/flask-express-cicd-deployment.git
```

✅ Installed Dependencies for Flask:

```
cd terraform-aws-flask-express-deployment/flask-backend
pip3 install --break-system-packages -r requirements.txt
```

✅ Installed Dependencies for Express:

```
cd ../express-frontend
npm install
```

4. Configured Applications

Flask Backend (app.py):

Runs on port 5000

Returns a JSON response:

```
{"message": "Hello from Flask backend!"}
```

Express Frontend (index.js):

Runs on port 3000

Displays frontend message and fetches data from Flask backend.

5. Run Applications Using PM2

To ensure both applications run continuously:

Start Flask

```
pm2 start app.py --name flask-backend --interpreter python3
```

Start Express

```
pm2 start index.js --name express-frontend
```

Save process list and enable startup on reboot

```
pm2 save
```

```
pm2 startup
```

Verification:

```
pm2 list
```

Both processes appeared as:

id	name	status	cpu	mem	user	watching

0	flask-backend	online	0%	3MB	ubuntu	disabled
1	express-front	online	0%	5MB	ubuntu	disabled

6. Application Verification

Flask accessible at:

👉 <http://<EC2-Public-IP>:5000>

Output:

```
{"message": "Hello from Flask backend!"}
```

Express accessible at:

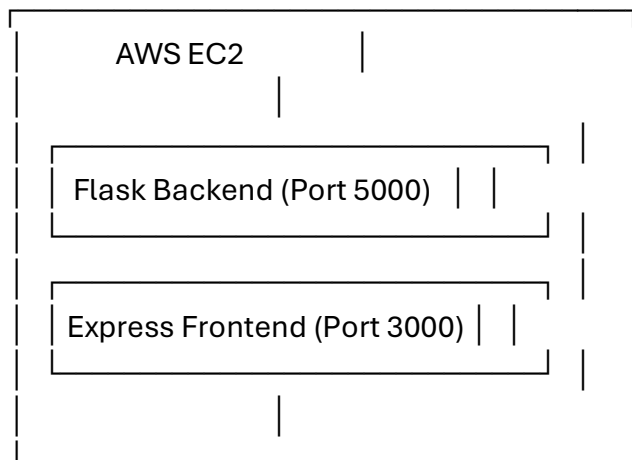
👉 <http://<EC2-Public-IP>:3000>

Output:

Express Frontend

Hello from Flask backend!

7. Architecture Diagram



✅ Deliverables for Part 1

Flask & Express both running on EC2 instance.

Applications accessible via public IPs.

Managed using PM2 for reliability.

Architecture and workflow documented.

PART 2: Implement CI/CD Pipeline Using Jenkins

🎯 Objective

Automate the deployment process for Flask and Express using Jenkins pipelines.

1. Jenkins Installation and Configuration

Installed Jenkins on the same EC2 instance:

```
curl -fsSL https://pkg.jenkins.io/debian/jenkins.io-2023.key | sudo tee
/usr/share/keyrings/jenkins-keyring.asc > /dev/null
echo "deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc]
https://pkg.jenkins.io/debian-stable binary/" | sudo tee /etc/apt/sources.list.d/jenkins.list
> /dev/null
sudo apt update -y
sudo apt install jenkins -y
sudo systemctl enable jenkins
sudo systemctl start jenkins
```

Access Jenkins via:

👉 <http://<EC2-Public-IP>:8080>

Installed required plugins:

Git Plugin

NodeJS Plugin

Python Plugin

Pipeline Plugin

2. Jenkins Pipeline Configuration

Flask Pipeline - Flask-Backend-CI-CD

Jenkinsfile:

```
pipeline {
  agent any
  stages {
    stage('Checkout Code') {
      steps {
        git branch: 'main', url: 'https://github.com/0079567603568sahal/flask-express-
cicd-deployment.git'
      }
    }
    stage('Install Dependencies') {
      steps {
        dir('terraform-aws-flask-express-deployment/flask-backend') {
          sh '''
          echo "Installing Flask dependencies..."
          python3 -m pip install --break-system-packages -r requirements.txt
          '''
        }
      }
    }
    stage('Run Flask Server') {
      steps {
        dir('terraform-aws-flask-express-deployment/flask-backend') {
          sh '''
          echo "Starting Flask backend..."
          pm2 delete flask-backend || true
          pm2 start app.py --name flask-backend --interpreter python3
          pm2 save
          '''
        }
      }
    }
  }
}
```

```

post {
    success {
        echo '✅ Flask backend deployed successfully!'
    }
    failure {
        echo '❌ Deployment failed.'
    }
}
}

```

Express Pipeline - Express-Frontend-CI-CD

Jenkinsfile:

```

pipeline {
    agent any
    stages {
        stage('Checkout Code') {
            steps {
                git branch: 'main', url: 'https://github.com/0079567603568sahal/flask-express-
cicd-deployment.git'
            }
        }
        stage('Install Dependencies') {
            steps {
                dir('terraform-aws-flask-express-deployment/express-frontend') {
                    sh '''
                    echo "Installing Express dependencies..."
                    npm install
                    '''
                }
            }
        }
        stage('Run Express Server') {
            steps {
                dir('terraform-aws-flask-express-deployment/express-frontend') {
                    sh '''

```

```

        echo "Starting Express frontend..."
        pm2 delete express-frontend || true
        pm2 start index.js --name express-frontend
        pm2 save
    ""
}
}
}
}
post {
    success {
        echo '✅ Express frontend deployed successfully!'
    }
    failure {
        echo '❌ Deployment failed.'
    }
}
}
}

```

3. Triggering the Pipeline

Configured GitHub Webhooks:

Added webhook in repository settings.

URL: `http://<EC2-Public-IP>:8080/github-webhook/`

Content type: `application/json`

Each code push triggers an automated Jenkins build.

4. Verification and Results

Jenkins Build Output:

✅ Successful pipeline stages for both Flask and Express.

Docker/PM2 Logs:

Both containers and apps confirmed running.

Browser Access:

Flask: <http://3.25.76.39:5000> → “Hello from Flask backend!”

Express: <http://3.25.76.39:3000> → “Express Frontend + Backend data.”

✅ Deliverables for Part 2

Jenkins installed and configured on EC2.

Two pipelines implemented for Flask and Express.

Code auto-pulled from GitHub via SCM.

Builds automatically deploy apps with PM2.

Successful Jenkins build evidence attached.

Final Outcome

- ✓ Flask backend and Express frontend successfully deployed.
- ✓ Jenkins CI/CD pipeline implemented and verified.
- ✓ Automated code deployment working on every commit.
- ✓ Both applications running live on AWS EC2.
- ✓ Docker/PM2 setup ensures persistence and reliability.

📁 Evidence / Screenshots

Instance Configuration

Jenkins URL:

The Jenkins URL is used to provide the root URL for absolute links to various Jenkins resources. That means this value is required for proper operation of many Jenkins features including email notifications, PR status updates, and the BUILD_URL environment variable provided to build steps.

The proposed default value shown is **not saved yet** and is generated from the current request, if possible. The best practice is to set this value to the URL that users are expected to use. This will avoid confusion when sharing or viewing links.

Q Git plugin				Install	
Install	Name	Released	Health		
☐	Git Push 34.vd474e0fe7b_ec git Source Code Management This plugin allows to perform a git push as a post build step via Git Plugin	3 yr 8 mo ago	85		
☐	Google Authenticated Source 0.4 Source Code Management related Source Code Management This plugin exposes a credential for use with the Git plugin for authenticating with Google source code hosting as a service account.	6 yr 0 mo ago	25		
☐	AWS CodeCommit URL Helper 0.0.3 Source Code Management related AWS CodeCommit Helper augments the Git Plugin in order to generate Credentials, without the need for the <code>aws-codecommit-credential-helper</code> command	7 yr 2 mo ago	76		
☐	Alternative build chooser 1.1 Source Code Management related Adds a new build choosing strategy to the Jenkins GIT plugin . Treats the configured branches as a list of possible branch names to try: build the first branch to match.	12 yr ago	64		
☐	Git PreBuildMerge Trait 1.0 Adds PreBuildMerge Trait to Git Plugin	8 yr 2 mo ago	61		

Q pipeline				Install	
Install	Name	Released	Health		
☐	Pipeline Graph Analysis 245.v88f03631a_b_21 Library plugins (for use by other plugins) Provides a REST API to access pipeline and pipeline run data.	1 mo 15 days ago	97		
☐	Pipeline: REST API 2.38 User Interface Provides a REST API to access pipeline and pipeline run data.	5 mo 28 days ago	100		
☐	Pipeline: Stage View 2.38 User Interface Pipeline Stage View Plugin.	5 mo 28 days ago	100		
☐	Docker Pipeline 634.vcdc7242b_eda_7 pipeline DevOps Deployment docker Build and use Docker containers from pipelines.	11 days ago	96		
☐	Lockable Resources 1438.v3c0f8c9e2060 pipeline Cluster Management Agent Management This plugin allows to define external resources (such as printers, phones, computers) that can be locked by builds. If a build requires an external resource which is already locked, it will wait for the resource to be free.	1 day 21 hr ago	100		
☐	Pipeline: Deprecated Groovy Libraries 615.vb_b_0664a_b_19f3 Miscellaneous Hosting of Pipeline Groovy libraries inside a Jenkins Git server. Deprecated. Use Pipeline: Groovy Libraries instead. If you see this plugin installed just because you upgraded, you can probably uninstall it now. This plugin should only be used if you have historically <i>pushed</i> libraries to a Git server inside Jenkins.	7 mo 17 days ago	68		

This plugin is deprecated. In general, this means that it is either obsolete, no longer being developed, or may no longer work. [Learn more.](#)

Plugins

- Updates
- Available plugins
- Installed plugins
- Advanced settings
- Download progress

Download progress

Preparation	- Checking internet connectivity - Checking update center connectivity - Success
Pyenv Pipeline	Success
ShiningPanda	Pending
Loading plugin extensions	Pending
Restarting Jenkins	Pending

- [Go back to the top page](#)
(you can start using the installed plugins right away)
- ☒ Restart Jenkins when installation is complete and no jobs are running

NodeJS

Name

Node18

☒ Install automatically ?

Install from nodejs.org

Version

NodeJS 25.0.0

For the underlying architecture, if available, force the installation of the 32bit package. Otherwise the build will fail

☐ Force 32bit architecture

Global npm packages to install

Specify list of packages to install globally -- see npm install -g. Note that you can fix the packages version by using the syntax 'packageName@version'

Global npm packages refresh hours

Duration, in hours, before 2 npm cache update. Note that 0 will always update npm cache

72

+ Add Installer

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Setting up node-yp (9.3.0-3) ...
Setting up node-traces (3.0.2+~3.0.1-1) ...
Setting up node-chokidar (3.6.0-2) ...
Setting up node-watchpack (2.4.0+~cs2.8.1-1) ...
Setting up node-micromatch (4.0.5+~4.0.2-1) ...
Setting up node-globby (13.1.3+~cs16.25.40-2) ...
Setting up node-dal (2.0.0-5) ...
Setting up node-find-cache-dir (3.3.2+~3.2.1-1) ...
Setting up node-istanbul (0.4.5+repack10+~cs98.25.59-2) ...
Setting up node-tape (5.6.1+~cs8.20.19-1) ...
Setting up node-babel (7.20.15+~cs214.269.168-6build1) ...
update-alternatives: using /usr/bin/babeljs-7 to provide /usr/bin/babeljs (babeljs) in auto mode
update-alternatives: using /usr/bin/babeljs-7-external-helpers to provide /usr/bin/babeljs-external-helpers (babeljs-external-helpers) in auto mode
update-alternatives: using /usr/bin/babeljs-7-node to provide /usr/bin/babeljs-node (babeljs-node) in auto mode
update-alternatives: using /usr/bin/babeljs-7-parser to provide /usr/bin/babeljs-parser (babeljs-parser) in auto mode
Setting up node-deep-equal (2.2.3+~cs43.15.94-1) ...
Setting up libjs-uglify (0.12.5+~1.0.10-1) ...
Setting up node-babel-plugin-lodash (3.3.4+~cs2.0.1-7) ...
Setting up node-jest-debbundle (29.6.2+~cs73.45.28-5) ...
Setting up node-tap (16.3.7+~cs50.9.19-4) ...
Setting up node-util (0.12.5+~1.0.10-1) ...
Setting up webpack (5.76.1+~git~cs17.16.16-1) ...
Setting up node-assert (2.0.0+~cs3.9.8-2) ...
Setting up node-css-loader (6.8.1+~cs14.0.17-1) ...
Setting up node-parse-json (5.2.0+~cs5.1.7-1) ...
Setting up npm (9.2.0+~cs1-2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
Processing triggers for man-db (2.12.0-4build1) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Pending kernel upgrade!
Running kernel version:
6.14.0-1011-aws
Diagnostics:
The currently running kernel version is not the expected kernel version 6.14.0-1015-aws.

Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.

Restarting services...

Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service
systemctl restart getty@tty1.service
systemctl restart networkd-dispatcher.service
systemctl restart serial-getty@ttyS0.service
systemctl restart systemd-journal.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

No user sessions are running outdated binaries.
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
6.0.13
ubuntu@ip-172-31-47-117:~$ git --version
git version 2.43.0
ubuntu@ip-172-31-47-117:~$ sudo apt install -y git
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
git is already the newest version (1:2.43.0-1ubuntu7.3).
0 upgraded, 0 newly installed, 0 to remove and 23 not upgraded.
ubuntu@ip-172-31-47-117:~$ cd -
git clone https://github.com/00795676835685ahal/terraform-aws-flask-express-deployment.git
cd terraform-aws-flask-express-deployment
Cloning into 'terraform-aws-flask-express-deployment'...
remote: Enumerating objects: 118, done.
remote: Counting objects: 100% (118/118), done.
remote: Compressing objects: 100% (109/109), done.
remote: Total 118 (delta 10), reused 46 (delta 7), pack-reused 0 (from 0)
Receiving objects: 100% (118/118), 6.24 MiB | 17.41 MiB/s, done.
Resolving deltas: 100% (10/10), done.
ubuntu@ip-172-31-47-117:~/terraform-aws-flask-express-deployment$ ls
README.md  'Terraform AWS ECS Fargate Flask + Express Deployment.pdf'  es  part1_single_ec2  part2_separate_ec2  scripts  yes
README.md.txt  apps  nohup.out  part1_single_ec2  part3_ecr_ecs  'terraform screen shot'
pip install -r requirements.txt
nohup python3 app.py &
-bash: cd: flask-backend: No such file or directory
error: externally-managed-environment

x This environment is externally managed
-> To install Python packages system-wide, try apt install
python3-xyz, where xyz is the package you are trying to
install.

If you wish to install a non-Debian-packaged Python package,
create a virtual environment using python3 -m venv path/to/venv.
Then use path/to/venv/bin/python and path/to/venv/bin/pip. Make
sure you have python3-full installed.

If you wish to install a non-Debian packaged Python application,
it may be easiest to use pipx install xyz, which will manage a
virtual environment for you. Make sure you have pipx installed.

See /usr/share/doc/python3.12/README.venv for more information.

note: If you believe this is a mistake, please contact your Python installation or OS distribution provider. You can override this, at the risk of breaking your Python installation or OS, by passing --break-system-packages.
hint: See PEP 668 for the detailed specification.
[1] 68340
ubuntu@ip-172-31-47-117:~/terraform-aws-flask-express-deployment$ nohup: ignoring input and appending output to 'nohup.out'
ls
README.md  'Terraform AWS ECS Fargate Flask + Express Deployment.pdf'  es  part1_single_ec2  part3_ecr_ecs  'terraform screen shot'
README.md.txt  apps  nohup.out  part2_separate_ec2  scripts  yes
[1]+ Exit 2 nohup python3 app.py
ubuntu@ip-172-31-47-117:~/terraform-aws-flask-express-deployment$ cd apps/flask_app
ubuntu@ip-172-31-47-117:~/terraform-aws-flask-express-deployment/apps/flask_app$ ls
Dockerfile  app.py  package-lock.json  package.json  requirements.txt
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
Restarting services...

Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service
systemctl restart getty@tty1.service
systemctl restart networkd-dispatcher.service
systemctl restart serial-getty@tty0.service
systemctl restart system-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-47-117:~/terraform-aws-flask-express-deployment/apps/flask_app$ python3 -m venv venv
source venv/bin/activate
(venv) ubuntu@ip-172-31-47-117:~/terraform-aws-flask-express-deployment/apps/flask_app$ pip install -r requirements.txt
Collecting Flask==2.2.5 (from -r requirements.txt (line 1))
  Downloading Flask-2.2.5-py3-none-any.whl.metadata (3.9 kB)
Collecting Werkzeug>2.2.2 (from Flask==2.2.5->-r requirements.txt (line 1))
  Downloading Werkzeug-3.1.3-py3-none-any.whl.metadata (3.7 kB)
Collecting Jinja2>3.0 (from Flask==2.2.5->-r requirements.txt (line 1))
  Downloading Jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting itsdangerous>2.0 (from Flask==2.2.5->-r requirements.txt (line 1))
  Downloading itsdangerous-2.2.0-py3-none-any.whl.metadata (1.9 kB)
Collecting click<8.3.0-py3-none-any.whl.metadata (2.6 kB)
  Downloading click-8.3.0-py3-none-any.whl.metadata (2.6 kB)
Collecting MarkupSafe>2.0 (from Jinja2>3.0->Flask==2.2.5->-r requirements.txt (line 1))
  Downloading MarkupSafe-3.0.3-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (2.7 kB)
Downloaded Flask-2.2.5-py3-none-any.whl (101 kB)
101.3/101.8 kB 6.2 MB/s eta 0:00:00
Downloaded click-8.3.0-py3-none-any.whl (107 kB)
107.3/107.3 kB 9.4 MB/s eta 0:00:00
Downloaded itsdangerous-2.2.0-py3-none-any.whl (16 kB)
Downloaded Jinja2-3.1.6-py3-none-any.whl (134 kB)
134.9/134.9 kB 12.3 MB/s eta 0:00:00
Downloaded Werkzeug-3.1.3-py3-none-any.whl (224 kB)
224.5/224.5 kB 19.3 MB/s eta 0:00:00
Downloaded MarkupSafe-3.0.3-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (22 kB)
Installing collected packages: MarkupSafe, itsdangerous, Click, Werkzeug, Jinja2, Flask
Successfully installed Flask-2.2.5 Jinja2-3.1.6 MarkupSafe-3.0.3 Werkzeug-3.1.3 click-8.3.0 itsdangerous-2.2.0
(venv) ubuntu@ip-172-31-47-117:~/terraform-aws-flask-express-deployment/apps/flask_app$ pip install --break-system-packages -r requirements.txt
Requirement already satisfied: Flask==2.2.5 in ./venv/lib/python3.12/site-packages (from -r requirements.txt (line 1)) (2.2.5)
Requirement already satisfied: Werkzeug>2.2.2 in ./venv/lib/python3.12/site-packages (from Flask==2.2.5->-r requirements.txt (line 1)) (3.1.3)
Requirement already satisfied: Jinja2>3.0 in ./venv/lib/python3.12/site-packages (from Flask==2.2.5->-r requirements.txt (line 1)) (3.1.6)
Requirement already satisfied: itsdangerous>2.0 in ./venv/lib/python3.12/site-packages (from Flask==2.2.5->-r requirements.txt (line 1)) (2.2.0)
Requirement already satisfied: click<8.0 in ./venv/lib/python3.12/site-packages (from Flask==2.2.5->-r requirements.txt (line 1)) (8.3.0)
Requirement already satisfied: MarkupSafe>2.0 in ./venv/lib/python3.12/site-packages (from Jinja2>3.0->Flask==2.2.5->-r requirements.txt (line 1)) (3.0.3)
(venv) ubuntu@ip-172-31-47-117:~/terraform-aws-flask-express-deployment/apps/flask_app$ nohup python3 app.py &
[1] 68624
(venv) ubuntu@ip-172-31-47-117:~/terraform-aws-flask-express-deployment/apps/flask_app$ ps aux | grep app.py
ps aux | grep app.py
ubuntu 68624 2.2 3.2 40528 30852 pts/0 S 11:43 0:00 python3 app.py
ubuntu 68626 0.0 0.2 7080 2080 pts/0 S+ 11:43 0:00 grep --color=auto app.py

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
k-express-deployment/apps/express_app$ nohup: ignoring input and appending output to 'nohup.out'
ps aux | grep node
ps aux | grep node
root 14599 0.0 1.8 11157172 17472 ? Ssl Oct16 0:04 node server.js
ubuntu 68747 4.4 6.8 619500 64592 pts/0 Sl 11:50 0:00 node index.js
ubuntu 68758 0.0 0.2 7076 2084 pts/0 S+ 11:50 0:00 grep --color=auto node
(venv) ubuntu@ip-172-31-47-117:~/terraform-aws-flask-express-deployment/apps/express_app$ curl http://localhost:3000
curl http://localhost:5000
Hello from Express App on port 3000{"message":"Hello from Flask running in Docker on port 5000!"}
(venv) ubuntu@ip-172-31-47-117:~/terraform-aws-flask-express-deployment/apps/express_app$

(venv) ubuntu@ip-172-31-47-117:~/terraform-aws-flask-express-deployment/apps/express_app$ nohup: ignoring input and appending output to 'nohup.out'
sudo ls -l :3000
sudo ls -l :3000
COWARD PID USER FD TYPE DEVICE SIZE/OFF NODE NAME
docker-pr-14636 root 7u IPv4 71831 0t0 TCP *:3000 (LISTEN)
docker-pr-14641 root 7u IPv6 71832 0t0 TCP *:3000 (LISTEN)
[2]+ Exit 1 nohup node index.js
(venv) ubuntu@ip-172-31-47-117:~/terraform-aws-flask-express-deployment/apps/express_app$ sudo kill -9 <PID>
bash: syntax error near unexpected token `newline'
(venv) ubuntu@ip-172-31-47-117:~/terraform-aws-flask-express-deployment/apps/express_app$ nohup node index.js &
[2] 68718
k-express-deployment/apps/express_app$ nohup: ignoring input and appending output to 'nohup.out'
ps aux | grep node
ps aux | grep node
root 14599 0.0 1.8 11157172 17472 ? Ssl Oct16 0:04 node server.js
ubuntu 68747 4.4 6.8 619500 64592 pts/0 Sl 11:50 0:00 node index.js
ubuntu 68758 0.0 0.2 7076 2084 pts/0 S+ 11:50 0:00 grep --color=auto node
(venv) ubuntu@ip-172-31-47-117:~/terraform-aws-flask-express-deployment/apps/express_app$ curl http://localhost:3000
curl http://localhost:5000
Hello from Express App on port 3000{"message":"Hello from Flask running in Docker on port 5000!"}
(venv) ubuntu@ip-172-31-47-117:~/terraform-aws-flask-express-deployment/apps/express_app$
```

- OAuth Apps
- Personal access tokens
- Fine-grained tokens
- Tokens (classic)

New personal access token (classic)

Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to [authenticate to the API over Basic Authentication](#).

Note

What's this token for?

Expiration

30 days (Nov 26, 2025)

The token will expire on the selected date

Select scopes

Scopes define the access for personal tokens. [Read more about OAuth scopes](#).

☐ repo	Full control of private repositories
☐ repo:status	Access commit status
☐ repo:deployment	Access deployment status
☐ public_repo	Access public repositories
☐ repo:invite	Access repository invitations
☐ security_events	Read and write security events
☒ workflow	Update GitHub Action workflows
☐ write:packages	Upload packages to GitHub Package Registry
☐ read:packages	Download packages from GitHub Package Registry
☐ delete:packages	Delete packages from GitHub Package Registry
☐ admin:org	Full control of orgs and teams, read and write org projects
☐ write:org	Read and write org and team membership, read and write org projects
☐ read:org	Read org and team membership, read org projects
☐ manage_runners:org	Manage org runners and runner groups
☐ admin:public_key	Full control of user public keys
☐ write:public_key	Write user public keys
☐ read:public_key	Read user public keys
☒ admin:repo_hook	Full control of repository hooks
☐ write:repo_hook	Write repository hooks



Jenkins installation fix x 3.25.76.39:5000 x Flask-Backend-CI-CD #21 flask-express-cicd-deploy 307996760256baha/fla flask-express-cicd-deploy Instance details | EC2 | sp Jenkins CI/CD Pipeline - G DEVOPS Lecture Page

< > Not secure 3.25.76.39:5000

Pretty print

{"message": "Hello from Flask backend!"}

New Item

Enter an item name

Express-Frontend-CI-CD

Select an item type



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.



Pipeline

Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.



Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.



Folder

Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.



Multibranch Pipeline

Creates a set of Pipeline projects according to detected branches in one SCM repository.



Organization Folder

Creates a set of multibranch project subfolders by scanning for repositories.

If you want to create a new item from other existing, you can use this option:

Copy from

OK

Status



#1 (27 Oct 2025, 08:59:07)

Add description

Keep this build forever

</> Changes



Started by user MUHAMMED SAHAL SR

Console Output



This run spent:

- 18 ms waiting
- 17 sec build duration
- 17 sec total from scheduled to completion.

Edit Build Information

Delete build #1

Timings

Git Build Data



Revision: 09ac046674b1194b9a66d91431082ff12c833ca8

Pipeline Overview

Repository: <https://github.com/0075567603566sahal/task-express-cicd-deployment.git>

Restart from Stage

- refs/remotes/origin/main

Replay



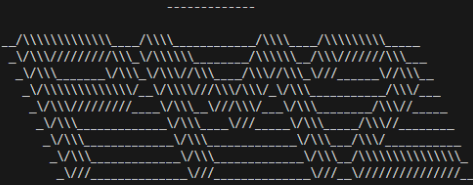
No changes.

Pipeline Steps

Workspaces

Started 20 sec ago
Took 17 sec

ubuntu@ip-172-31-40-34:~\$ pm2 list



Runtime Edition

PM2 is a Production Process Manager for Node.js applications
with a built-in Load Balancer.

Start and Daemonize any application:
\$ pm2 start app.js

Load Balance 4 instances of api.js:
\$ pm2 start api.js -i 4

Monitor in production:
\$ pm2 monitor

Make pm2 auto-boot at server restart:
\$ pm2 startup

To go further checkout:
<http://pm2.io/>

[PM2] Spawning PM2 daemon with pm2_home=/home/ubuntu/.pm2
[PM2] PM2 Successfully daemonized

id	name	namespace	version	mode	pid	uptime	0	status	cpu	mem	user	watching
----	------	-----------	---------	------	-----	--------	---	--------	-----	-----	------	----------

ubuntu@ip-172-31-40-34:~\$

Last login: Mon Oct 27 06:56:59 2025 from 111.92.27.39
ubuntu@ip-172-31-40-34:~\$ pm2 list



Runtime Edition

PM2 is a Production Process Manager for Node.js applications
with a built-in Load Balancer.

Start and Daemonize any application:
\$ pm2 start app.js

Load Balance 4 instances of api.js:
\$ pm2 start api.js -i 4

Monitor in production:
\$ pm2 monitor

Make pm2 auto-boot at server restart:
\$ pm2 startup

To go further checkout:
<http://pm2.io/>

[PM2] Spawning PM2 daemon with pm2_home=/home/ubuntu/.pm2
[PM2] PM2 Successfully daemonized

id	name	namespace	version	mode	pid	uptime	0	status	cpu	mem	user	watching
----	------	-----------	---------	------	-----	--------	---	--------	-----	-----	------	----------

```
ubuntu@ip-172-31-40-34:~$ pm2 list
```

id	name	namespace	version	mode	pid	uptime	0	status	cpu	mem	user	watching
----	------	-----------	---------	------	-----	--------	---	--------	-----	-----	------	----------

```
ubuntu@ip-172-31-40-34:~$
```

```
ubuntu@ip-172-31-40-34:/var/lib/jenkins/workspace/Flask-Backend-CI-CD/terraform-aws-flask-express-deployment/Flask-backend$ python3 -m pip install --break-system-packages -r requirements.txt
Defaulting to user installation because normal site-packages is not writable
Collecting Flask==2.3.3 (from -r requirements.txt (line 1))
  Downloading flask-2.3.3-py3-none-any.whl.metadata (3.6 kB)
Collecting gunicorn==21.2.0 (from -r requirements.txt (line 2))
  Downloading gunicorn-21.2.0-py3-none-any.whl.metadata (4.1 kB)
Collecting Werkzeug==2.3.7 (from Flask==2.3.3->-r requirements.txt (line 1))
  Using cached werkzeug-3.1.3-py3-none-any.whl.metadata (3.7 kB)
Requirement already satisfied: Jinja2>=3.1.2 in /usr/lib/python3/dist-packages (from Flask==2.3.3->-r requirements.txt (line 1)) (3.1.2)
Collecting itsdangerous>=2.1.2 (from Flask==2.3.3->-r requirements.txt (line 1))
Requirement already satisfied: click>=8.1.3 in /usr/lib/python3/dist-packages (from Flask==2.3.3->-r requirements.txt (line 1)) (8.1.6)
Requirement already satisfied: blinker>=1.6.2 in /usr/lib/python3/dist-packages (from Flask==2.3.3->-r requirements.txt (line 1)) (1.7.0)
Requirement already satisfied: packaging in /usr/lib/python3/dist-packages (from gunicorn==21.2.0->-r requirements.txt (line 2)) (24.0)
Requirement already satisfied: MarkupSafe>=2.1.1 in /usr/lib/python3/dist-packages (from Werkzeug==2.3.7->Flask==2.3.3->-r requirements.txt (line 1)) (2.1.5)
Downloading flask-2.3.3-py3-none-any.whl (96 kB)
   96.1/96.1 kB 4.2 MB/s eta 0:00:00
Downloading gunicorn-21.2.0-py3-none-any.whl (80 kB)
   80.2/80.2 kB 7.5 MB/s eta 0:00:00
Using cached itsdangerous-2.2.0-py3-none-any.whl (16 kB)
Using cached werkzeug-3.1.3-py3-none-any.whl (224 kB)
Installing collected packages: Werkzeug, itsdangerous, gunicorn, Flask
  WARNING: The script gunicorn is installed in /home/ubuntu/.local/bin which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
  WARNING: The script flask is installed in /home/ubuntu/.local/bin which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed Flask-2.3.3 Werkzeug-3.1.3 gunicorn-21.2.0 itsdangerous-2.2.0
ubuntu@ip-172-31-40-34:/var/lib/jenkins/workspace/Flask-Backend-CI-CD/terraform-aws-flask-express-deployment/Flask-backend$ python3 -m flask --version
Python 3.12.3
Flask 2.3.3
Werkzeug 3.1.3
ubuntu@ip-172-31-40-34:/var/lib/jenkins/workspace/Flask-Backend-CI-CD/terraform-aws-flask-express-deployment/Flask-backend$ python3 app.py
 * Serving Flask app 'app'
 * Debug mode: off
Address already in use
Port 5000 is in use by another program. Either identify and stop that program, or start the server with a different port.
ubuntu@ip-172-31-40-34:/var/lib/jenkins/workspace/Flask-Backend-CI-CD/terraform-aws-flask-express-deployment/Flask-backend$ pm2 start app.py --name flask-backend --interpreter python3
pm2 save
[PM2] Starting /var/lib/jenkins/workspace/Flask-Backend-CI-CD/terraform-aws-flask-express-deployment/Flask-backend/app.py in fork_mode (1 instance)
[PM2] Done.
```

id	name	namespace	version	mode	pid	uptime	0	status	cpu	mem	user	watching
0	flask-backend	default	N/A	fork	2376	0s	0	online	0%	3.0mb	ubuntu	disabled

```
[PM2] Saving current process list...
[PM2] Successfully saved in /home/ubuntu/.pm2/dump.pm2
```

```
ubuntu@ip-172-31-40-34:/var/lib/jenkins/workspace/Flask-Backend-CI-CD/terraform-aws-flask-express-deployment/Flask-backend$ sudo ss -l -t :5000
COMMAND PID USER FD TYPE DEVICE SIZE/OFF NODE NAME
docker-pr 1375 root 7u IPv4 8583 0t0 TCP *:5000 (LISTEN)
docker-pr 1381 root 7u IPv6 8584 0t0 TCP *:5000 (LISTEN)
ubuntu@ip-172-31-40-34:/var/lib/jenkins/workspace/Flask-Backend-CI-CD/terraform-aws-flask-express-deployment/Flask-backend$ sudo docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
def24d3ae349 aws-flask-express-deploy_backend "gunicorn -w 3 -b 0.0.0.0:..." 21 minutes ago Up 21 minutes 0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp aws-flask-express-deploy_backend_1
b8c6ee707ffe aws-flask-express-deploy_frontend "docker-entrypoint.sh" 21 minutes ago Up 21 minutes 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp aws-flask-express-deploy_frontend_1
```

```
ubuntu@ip-172-31-40-34:/var/lib/jenkins/workspace/Flask-Backend-CI-CD/terraform-aws-flask-express-deployment/Flask-backend$ sudo docker logs def24d3ae349 --tail 20
[2025-10-27 08:52:38 +0000] [1] [Info] Starting gunicorn 23.0.0
[2025-10-27 08:52:38 +0000] [1] [Info] Listening at: http://0.0.0.0:5000 (1)
[2025-10-27 08:52:38 +0000] [1] [Info] Using worker: sync
[2025-10-27 08:52:38 +0000] [7] [Info] Booting worker with pid: 7
[2025-10-27 08:52:38 +0000] [8] [Info] Booting worker with pid: 8
[2025-10-27 08:52:38 +0000] [9] [Info] Booting worker with pid: 9
Express server running on port 3000
```

```
ubuntu@ip-172-31-40-34:/var/lib/jenkins/workspace/Flask-Backend-CI-CD/terraform-aws-flask-express-deployment/Flask-backend$ sudo docker ps --format 'table {{.ID}}\t{{.Names}}\t{{.Status}}'
CONTAINER ID NAMES
def24d3ae349 aws-flask-express-deploy_backend_1 Up 23 minutes
b8c6ee707ffe aws-flask-express-deploy_frontend_1 Up 23 minutes
```

```
ubuntu@ip-172-31-40-34:/var/lib/jenkins/workspace/Flask-Backend-CI-CD/terraform-aws-flask-express-deployment/Flask-backend$ sudo docker update --restart always def24d3ae349
sudo docker update --restart always b8c6ee707ffe
def24d3ae349
b8c6ee707ffe
```

```
Mohds@Mohdsahal MINGW64 ~/Downloads (main)
○ $ ssh -i "/c/Users/Mohds/Downloads/mohdsahal924 (1).pem" ubuntu@15.206.91.151
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1011-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Oct 25 05:01:46 UTC 2025

System load: 0.0           Temperature: -273.1 C
Usage of /: 31.2% of 6.71GB Processes: 113
Memory usage: 25%         Users logged in: 0
Swap usage: 0%            IPv4 address for ens5: 172.31.45.194

 * Ubuntu Pro delivers the most comprehensive open source security and
  compliance features.

  https://ubuntu.com/aws/pro

Expanded Security Maintenance for Applications is not enabled.

61 updates can be applied immediately.
38 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-45-194:~$ sudo apt update -y
sudo apt install -y python3 python3-pip git curl
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
55 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
```

```
fontconfig-config fonts-dejavu-core fonts-dejavu-mono g++ g++-13 g++-13-x86-64-linux-gnu g++-x86-64-linux-gnu gcc gcc-13 gcc-13-base gcc-13-x86-64-linux-gnu
gcc-x86-64-linux-gnu javascript-common libalgorithm-diff-perl libalgorithm-diff-xs-perl libalgorithm-merge-perl libaom3 libasan8 libatomic1 libbinutils libc-bin
libc-dev-bin libc-devtools libc6 libc6-dev libc6-i386 libcrypt-dev libctf-nobfd0 libctf0 libde265-0 libdeflate0 libdpkg-perl libexpat1-dev libfakeroot
libfile-fcntllock-perl libfontconfig1 libgcc-13-dev libgd3 libgomp1 libgprofng0 libheif-plugin-aomdec libheif-plugin-aomenc libheif-plugin-libde265 libheif1
libhwasan0 libisl23 libitm1 libjbig0 libjpeg-turbo8 libjpeg8 libjs-jquery libjs-sphinxdoc libjs-underscore liblerc4 liblsan0 libmpc3 libpython3-dev
libpython3.12-dev libpython3.12-minimal libpython3.12-stdlib libpython3.12t64 libquadmath0 libsqlite3-dev libstdc++-13-dev libtiff6 libubsan1 libubsan1-dev
libwebp7 libxpm4 linux-libc-dev linux-tools-common locales lto-disabled-list make manpages-dev python3-dev python3-wheel python3.12 python3.12-dev
python3.12-minimal rpcsvc-proto zlib1g-dev
Suggested packages:
  binutils-doc gprofng-gui bzip2-doc cpp-doc gcc-13-locales cpp-13-doc debsig-verify debian-keyring g++-multilib g++-13-multilib gcc-13-doc gcc-multilib autoconf
  automake libtool flex bison gdb gcc-doc gcc-13-multilib gdb-x86-64-linux-gnu apache2 | lighttpd | httpd glibc-doc libnss-nis libnss-nisplus bzip libgd-tools
  libheif-plugin-x265 libheif-plugin-fmpegdec libheif-plugin-jpegdec libheif-plugin-jpegenc libheif-plugin-j2kdec libheif-plugin-j2kenc libheif-plugin-rav1e
  libheif-plugin-svtenc libstdc++-13-doc make-doc python3.12-venv python3.12-doc binfmt-support
The following NEW packages will be installed:
  binutils binutils-common binutils-x86-64-linux-gnu build-essential bzip2 cpp cpp-13 cpp-13-x86-64-linux-gnu cpp-x86-64-linux-gnu dpkg-dev fakeroot
  fontconfig-config fonts-dejavu-core fonts-dejavu-mono g++ g++-13 g++-13-x86-64-linux-gnu g++-x86-64-linux-gnu gcc gcc-13 gcc-13-base gcc-13-x86-64-linux-gnu
  gcc-x86-64-linux-gnu javascript-common libalgorithm-diff-perl libalgorithm-diff-xs-perl libalgorithm-merge-perl libaom3 libasan8 libatomic1 libbinutils
  libc-dev-bin libc-devtools libc6-dev libc6-i386 libcrypt-dev libctf-nobfd0 libctf0 libde265-0 libdeflate0 libdpkg-perl libexpat1-dev libfakeroot
  libfile-fcntllock-perl libfontconfig1 libgcc-13-dev libgd3 libgomp1 libgprofng0 libheif-plugin-aomdec libheif-plugin-aomenc libheif-plugin-libde265 libheif1
  libhwasan0 libisl23 libitm1 libjbig0 libjpeg-turbo8 libjpeg8 libjs-jquery libjs-sphinxdoc libjs-underscore liblerc4 liblsan0 libmpc3 libpython3-dev
  libpython3.12-dev libquadmath0 libsqlite3-dev libstdc++-13-dev libtiff6 libubsan1 libubsan1-dev libwebp7 libxpm4 linux-libc-dev lto-disabled-list make
  manpages-dev python3-dev python3-pip python3-wheel python3.12-dev rpcsvc-proto zlib1g-dev
The following packages will be upgraded:
  dpkg libc-bin libc6 libpython3.12-minimal libpython3.12-stdlib libpython3.12t64 linux-tools-common locales python3.12 python3.12-minimal
10 upgraded, 86 newly installed, 0 to remove and 45 not upgraded.
Need to get 87.3 MB/106 MB of archives.
After this operation, 310 MB of additional disk space will be used.
Get:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 binutils-common amd64 2.42-4ubuntu2.5 [240 kB]
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libsqlite3-dev amd64 3.45.1-1ubuntu0.1 [15.5 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libbinutils amd64 2.42-4ubuntu2.5 [577 kB]
Get:4 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libctf-nobfd0 amd64 2.42-4ubuntu2.5 [97.7 kB]
Get:5 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libctf0 amd64 2.42-4ubuntu2.5 [94.5 kB]
Get:6 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libgprofng0 amd64 2.42-4ubuntu2.5 [849 kB]
Get:7 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 binutils-x86-64-linux-gnu amd64 2.42-4ubuntu2.5 [2462 kB]
Get:8 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 binutils amd64 2.42-4ubuntu2.5 [18.1 kB]
Get:9 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libc-dev-bin amd64 2.39-0ubuntu8.6 [20.4 kB]
Get:10 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-libc-dev amd64 6.8.0-86.87 [1932 kB]
Get:11 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libcrypt-dev amd64 1:4.4.36-4build1 [112 kB]
Get:12 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 rpcsvc-proto amd64 1.4.2-0ubuntu7 [67.4 kB]
Get:13 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libc6-dev amd64 2.39-0ubuntu8.6 [2125 kB]
Get:14 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 gcc-13-base amd64 13.3.0-6ubuntu2~24.04 [51.5 kB]
Get:15 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libisl23 amd64 0.26-3build1.1 [680 kB]
Get:16 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libmpc3 amd64 1.3.1-1build1.1 [54.6 kB]
Get:17 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 cpp-13-x86-64-linux-gnu amd64 13.3.0-6ubuntu2~24.04 [10.7 MB]
Get:18 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 cpp-13 amd64 13.3.0-6ubuntu2~24.04 [1038 B]
Get:19 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 cpp-x86-64-linux-gnu amd64 4:13.2.0-7ubuntu1 [5326 B]
Get:20 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 cpp amd64 4:13.2.0-7ubuntu1 [22.4 kB]
Get:21 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libc6-i386 amd64 2.39-0ubuntu8.6 [48.0 kB]
Get:22 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libgomp1 amd64 14.2.0-4ubuntu2~24.04 [148 kB]
Get:23 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libitm1 amd64 14.2.0-4ubuntu2~24.04 [29.7 kB]
Get:24 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libatomic1 amd64 14.2.0-4ubuntu2~24.04 [10.5 kB]
Get:25 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libasan8 amd64 14.2.0-4ubuntu2~24.04 [3031 kB]
Get:26 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 liblsan0 amd64 14.2.0-4ubuntu2~24.04 [1322 kB]
```

```
Get:54 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libalgorithm-diff-xs-perl amd64 0.04-8build3 [11.2 kB]
Get:55 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libalgorithm-merge-perl all 0.08-5 [11.4 kB]
Get:56 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libaom3 amd64 3.8.2-2ubuntu0.1 [1941 kB]
Get:57 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libfontconfig1 amd64 2.15.0-1.1ubuntu2 [139 kB]
Get:58 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libsharpv0 amd64 1.3.2-0.4build3 [15.8 kB]
Get:59 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libheif-plugin-aomdec amd64 1.17.6-1ubuntu4.1 [10.4 kB]
Get:60 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libde265-0 amd64 1.0.15-1build3 [166 kB]
Get:61 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libheif-plugin-libde265 amd64 1.17.6-1ubuntu4.1 [8176 B]
Get:62 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libheif1 amd64 1.17.6-1ubuntu4.1 [275 kB]
Get:63 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libjpeg-turbo8 amd64 2.1.5-2ubuntu2 [150 kB]
Get:64 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libjpeg8 amd64 8c-2ubuntu11 [2148 B]
Get:65 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libdeflate0 amd64 1.19-1build1.1 [43.9 kB]
Get:66 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libjbig0 amd64 2.1-6.1ubuntu2 [29.7 kB]
Get:67 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 liblerc4 amd64 4.0.0+ds-4ubuntu2 [179 kB]
Get:68 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libwebp7 amd64 1.3.2-0.4build3 [230 kB]
Get:69 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libtiff6 amd64 4.5.1+git230720-4ubuntu2.4 [199 kB]
Get:70 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libxpm4 amd64 1:3.5.17-1build2 [36.5 kB]
Get:71 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libgd3 amd64 2.3.3-9ubuntu5 [128 kB]
Get:72 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libc-devtools amd64 2.39-0ubuntu8.6 [29.3 kB]
Get:73 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libxpat1-dev amd64 2.6.1-2ubuntu0.3 [140 kB]
Get:74 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libfile-fcntllock-perl amd64 0.22-4ubuntu5 [30.7 kB]
Get:75 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libheif-plugin-aomenc amd64 1.17.6-1ubuntu4.1 [14.7 kB]
Get:76 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libjs-jquery all 3.6.1+dfsg+~3.5.14-1 [328 kB]
Get:77 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libjs-underscore all 1.13.4~dfsg+~1.11.4-3 [118 kB]
Get:78 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libjs-sphinxdoc all 7.2.6-6 [149 kB]
Get:79 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 zlib1g-dev amd64 1:1.3.dfsg-3.1ubuntu2.1 [894 kB]
Get:80 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libpython3.12-dev amd64 3.12.3-1ubuntu0.8 [5677 kB]
Get:81 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libpython3-dev amd64 3.12.3-0ubuntu2 [10.3 kB]
Get:82 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 manpages-dev all 6.7-2 [2013 kB]
Get:83 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 python3.12-dev amd64 3.12.3-1ubuntu0.8 [498 kB]
Get:84 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 python3-dev amd64 3.12.3-0ubuntu2 [26.7 kB]
Get:85 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 python3-wheel all 0.42.0-2 [53.1 kB]
Get:86 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 python3-pip all 24.0+dfsg-1ubuntu1.3 [1320 kB]
Fetched 87.3 MB in 1s (92.2 MB/s)
Extracting templates from packages: 100%
Preconfiguring packages ...
(Reading database ... 71718 files and directories currently installed.)
Preparing to unpack .../dpkg_1.22.6ubuntu6.5_amd64.deb ...
Unpacking dpkg (1.22.6ubuntu6.5) over (1.22.6ubuntu6.1) ...
Setting up dpkg (1.22.6ubuntu6.5) ...
(Reading database ... 71718 files and directories currently installed.)
Preparing to unpack .../libc6_2.39-0ubuntu8.6_amd64.deb ...
Unpacking libc6:amd64 (2.39-0ubuntu8.6) over (2.39-0ubuntu8.5) ...
Setting up libc6:amd64 (2.39-0ubuntu8.6) ...
(Reading database ... 71718 files and directories currently installed.)
Preparing to unpack .../libc-bin_2.39-0ubuntu8.6_amd64.deb ...
Unpacking libc-bin (2.39-0ubuntu8.6) over (2.39-0ubuntu8.5) ...
Setting up libc-bin (2.39-0ubuntu8.6) ...
(Reading database ... 71718 files and directories currently installed.)
Preparing to unpack .../00-libpython3.12t64_3.12.3-1ubuntu0.8_amd64.deb ...
Unpacking libpython3.12t64:amd64 (3.12.3-1ubuntu0.8) over (3.12.3-1ubuntu0.7) ...
Preparing to unpack .../01-python3.12_3.12.3-1ubuntu0.8_amd64.deb ...
Unpacking python3.12 (3.12.3-1ubuntu0.8) over (3.12.3-1ubuntu0.7) ...
```

```

ubuntu@ip-172-31-45-194:~/terraform-aws-flask-express-deployment/apps/flask_app$ sudo apt update -y
sudo apt install -y python3.12-venv
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 https://deb.nodesource.com/node_18.x nodistro InRelease
Hit:5 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
45 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  python3-pip-whl python3-setuptools-whl
The following NEW packages will be installed:
  python3-pip-whl python3-setuptools-whl python3.12-venv
0 upgraded, 3 newly installed, 0 to remove and 45 not upgraded.
Need to get 2429 kB of archives.
After this operation, 2777 kB of additional disk space will be used.
Get:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 python3-pip-whl all 24.0+dfsg-1ubuntu1.3 [1787 kB]
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 python3-setuptools-whl all 68.1.2-2ubuntu1.2 [716 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 python3.12-venv amd64 3.12.3-1ubuntu0.8 [5678 B]
Fetched 2429 kB in 0s (58.4 MB/s)
Selecting previously unselected package python3-pip-whl.
(Reading database ... 84479 files and directories currently installed.)
Preparing to unpack .../python3-pip-whl_24.0+dfsg-1ubuntu1.3_all.deb ...
Unpacking python3-pip-whl (24.0+dfsg-1ubuntu1.3) ...
Selecting previously unselected package python3-setuptools-whl.
Preparing to unpack .../python3-setuptools-whl_68.1.2-2ubuntu1.2_all.deb ...
Unpacking python3-setuptools-whl (68.1.2-2ubuntu1.2) ...
Selecting previously unselected package python3.12-venv.
Preparing to unpack .../python3.12-venv_3.12.3-1ubuntu0.8_amd64.deb ...
Unpacking python3.12-venv (3.12.3-1ubuntu0.8) ...
Setting up python3-setuptools-whl (68.1.2-2ubuntu1.2) ...
Setting up python3-pip-whl (24.0+dfsg-1ubuntu1.3) ...
Setting up python3.12-venv (3.12.3-1ubuntu0.8) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service
systemctl restart getty@tty1.service
systemctl restart networkd-dispatcher.service
systemctl restart serial-getty@ttyS0.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

ubuntu@ip-172-31-45-194:~/terraform-aws-flask-express-deployment/apps/flask_app$ cd ~/terraform-aws-flask-express-deployment/apps/Flask_app
python3 -m venv venv
source venv/bin/activate
pip install -r requirements.txt
Collecting Flask==2.2.5 (from -r requirements.txt (line 1))
  Downloading Flask-2.2.5-py3-none-any.whl.metadata (3.9 kB)
Collecting Werkzeug>=2.2.2 (from Flask==2.2.5->-r requirements.txt (line 1))
  Downloading werkzeug-3.1.3-py3-none-any.whl.metadata (3.7 kB)
Collecting Jinja2>=3.0 (from Flask==2.2.5->-r requirements.txt (line 1))
  Downloading jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting itsdangerous>=2.0 (from Flask==2.2.5->-r requirements.txt (line 1))
  Downloading itsdangerous-2.2.0-py3-none-any.whl.metadata (1.9 kB)
Collecting click>=8.0 (from Flask==2.2.5->-r requirements.txt (line 1))
  Downloading click-8.3.0-py3-none-any.whl.metadata (2.6 kB)
Collecting MarkupSafe>=2.0 (from Jinja2>=3.0->Flask==2.2.5->-r requirements.txt (line 1))
  Downloading markupsafe-3.0.3-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (2.7 kB)
Downloading Flask-2.2.5-py3-none-any.whl (101 kB)
 101.8/101.8 kB 6.2 MB/s eta 0:00:00
Downloading click-8.3.0-py3-none-any.whl (107 kB)
 107.3/107.3 kB 9.8 MB/s eta 0:00:00
Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
Downloading jinja2-3.1.6-py3-none-any.whl (134 kB)
 134.9/134.9 kB 13.3 MB/s eta 0:00:00
Downloading werkzeug-3.1.3-py3-none-any.whl (224 kB)
 224.5/224.5 kB 20.4 MB/s eta 0:00:00
Downloading markupsafe-3.0.3-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (22 kB)
Installing collected packages: MarkupSafe, itsdangerous, click, Werkzeug, Jinja2, Flask
Successfully installed Flask-2.2.5 Jinja2-3.1.6 MarkupSafe-3.0.3 Werkzeug-3.1.3 click-8.3.0 itsdangerous-2.2.0
(venv) ubuntu@ip-172-31-45-194:~/terraform-aws-flask-express-deployment/apps/flask_app$

```

```

ubuntu@ip-172-31-45-194:~/terraform-aws-flask-express-deployment/apps/flask_app$ cd ~/terraform-aws-flask-express-deployment/apps/flask_app
python3 -m venv venv
source venv/bin/activate
pip install -r requirements.txt
Collecting Flask==2.2.5 (from -r requirements.txt (line 1))
  Downloading Flask-2.2.5-py3-none-any.whl.metadata (3.9 kB)
Collecting Werkzeug>=2.2.2 (from Flask==2.2.5->-r requirements.txt (line 1))
  Downloading werkzeug-3.1.3-py3-none-any.whl.metadata (3.7 kB)
Collecting Jinja2>=3.0 (from Flask==2.2.5->-r requirements.txt (line 1))
  Downloading jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting itsdangerous>=2.0 (from Flask==2.2.5->-r requirements.txt (line 1))
  Downloading itsdangerous-2.2.0-py3-none-any.whl.metadata (1.9 kB)
Collecting click>=8.0 (from Flask==2.2.5->-r requirements.txt (line 1))
  Downloading click-8.3.0-py3-none-any.whl.metadata (2.6 kB)
Collecting MarkupSafe>=2.0 (from Jinja2>=3.0->Flask==2.2.5->-r requirements.txt (line 1))
  Downloading markupsafe-3.0.3-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (2.7 kB)
Downloading Flask-2.2.5-py3-none-any.whl (101 kB)
101.8/101.8 kB 6.2 MB/s eta 0:00:00
Downloading click-8.3.0-py3-none-any.whl (107 kB)
107.3/107.3 kB 9.8 MB/s eta 0:00:00
Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
Downloading Jinja2-3.1.6-py3-none-any.whl (134 kB)
134.9/134.9 kB 13.3 MB/s eta 0:00:00
Downloading werkzeug-3.1.3-py3-none-any.whl (224 kB)
224.5/224.5 kB 20.4 MB/s eta 0:00:00
Downloading markupsafe-3.0.3-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (22 kB)
Installing collected packages: MarkupSafe, itsdangerous, click, Werkzeug, Jinja2, Flask
Successfully installed Flask-2.2.5 Jinja2-3.1.6 MarkupSafe-3.0.3 Werkzeug-3.1.3 click-8.3.0 itsdangerous-2.2.0
(venv) ubuntu@ip-172-31-45-194:~/terraform-aws-flask-express-deployment/apps/flask_app$ python app.py
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.31.45.194:5000
Press CTRL+C to quit

```

```

^C(venv) ubuntu@ip-172-31-45-194:~/terraform-aws-flask-express-deployment/apps/flask_app$ pm2 start "venv/bin/python app.py" --name flask-app
pm2 save
pm2 startup
[PM2] Starting /usr/bin/bash in fork_mode (1 instance)
[PM2] Done.

```

id	name	namespace	version	mode	pid	uptime	0	status	cpu	mem	user	watching
0	flask-app	default	N/A	fork	6998	0s	0	online	0%	9.7mb	ubuntu	disabled

```

[PM2] Saving current process list...
[PM2] Successfully saved in /home/ubuntu/.pm2/dump.pm2
[PM2] Init System found: systemd
[PM2] To setup the Startup Script, copy/paste the following command:
sudo env PATH=$PATH:/usr/bin /usr/lib/node_modules/pm2/bin/pm2 startup systemd -u ubuntu --hp /home/ubuntu
(venv) ubuntu@ip-172-31-45-194:~/terraform-aws-flask-express-deployment/apps/flask_app$

```

```

(venv) ubuntu@ip-172-31-45-194:~/terraform-aws-flask-express-deployment/apps/flask_app$ pm2 list

```

id	name	namespace	version	mode	pid	uptime	0	status	cpu	mem	user	watching
0	flask-app	default	N/A	fork	6998	21s	0	online	0%	30.2mb	ubuntu	disabled

```

(venv) ubuntu@ip-172-31-45-194:~/terraform-aws-flask-express-deployment/apps/flask_app$

```

Pretty print ☐
 ("message": "Hello from Flask running in Docker on port 5000!")


```
(venv) ubuntu@ip-172-31-45-194:~/terraform-aws-flask-express-deployment/apps/flask_app$ cd ~/terraform-aws-flask-express-deployment/apps/express_app
npm install
node index.js

added 73 packages, and audited 74 packages in 2s

14 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
npm notice
npm notice New major version of npm available! 10.8.2 -> 11.6.2
npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.6.2
npm notice To update run: npm install -g npm@11.6.2
npm notice
Express app listening on port 3000
```

```
(venv) ubuntu@ip-172-31-45-194:~/terraform-aws-flask-express-deployment/apps/express_app$ pm2 start "node index.js" --name express-app
pm2 save
pm2 startup
[PM2] Starting /usr/bin/bash in fork_mode (1 instance)
[PM2] Done.
```

id	name	namespace	version	mode	pid	uptime	0	status	cpu	mem	user	watching
1	express-app	default	N/A	fork	7059	0s	0	online	0%	20.0mb	ubuntu	disabled
0	flask-app	default	N/A	fork	6998	3m	0	online	0%	30.2mb	ubuntu	disabled

```
[PM2] Saving current process list...
[PM2] Successfully saved in /home/ubuntu/.pm2/dump.pm2
[PM2] Init System found: systemd
[PM2] To setup the Startup Script, copy/paste the following command:
sudo env PATH=$PATH:/usr/bin:/usr/lib/node_modules/pm2/bin/pm2 startup systemd -u ubuntu --hp /home/ubuntu
```

< > C  Not secure 15.206.91.151:3000   

Hello from Express App on port 3000

```
(venv) ubuntu@ip-172-31-45-194:~/terraform-aws-flask-express-deployment/apps/express_app$ sudo ufw status
Status: inactive
(venv) ubuntu@ip-172-31-45-194:~/terraform-aws-flask-express-deployment/apps/express_app$ sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
   Active: active (running) since Sat 2025-10-25 05:32:32 UTC; 20min ago
     Main PID: 10699 (java)
       Tasks: 38 (limit: 1000)
      Memory: 288.6M (peak: 321.0M)
         CPU: 22.652s
    CGroup: /system.slice/jenkins.service
            └─10699 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/var --httpPort=8080

Oct 25 05:32:26 ip-172-31-45-194 jenkins[10699]: [LF]> This may also be found at: /var/lib/jenkins/secrets/InitialAdminPassword
Oct 25 05:32:26 ip-172-31-45-194 jenkins[10699]: [LF]>
Oct 25 05:32:26 ip-172-31-45-194 jenkins[10699]: [LF]> *****
Oct 25 05:32:26 ip-172-31-45-194 jenkins[10699]: [LF]> *****
Oct 25 05:32:26 ip-172-31-45-194 jenkins[10699]: [LF]> *****
Oct 25 05:32:26 ip-172-31-45-194 jenkins[10699]: [LF]> *****
Oct 25 05:32:32 ip-172-31-45-194 jenkins[10699]: 2025-10-25 05:32:32.240+0000 [id=30] INFO jenkins.InitReactorRunner$1onAttained: Completed initialization
Oct 25 05:32:32 ip-172-31-45-194 jenkins[10699]: 2025-10-25 05:32:32.269+0000 [id=23] INFO hudson.lifecycle.Lifecycle$onReady: Jenkins is fully up and running
Oct 25 05:32:32 ip-172-31-45-194 systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
Oct 25 05:32:34 ip-172-31-45-194 jenkins[10699]: 2025-10-25 05:32:34.286+0000 [id=49] INFO h.m.DownloadService$Downloadable#load: Obtained the updated data file for hudson.tasks.Maven.MavenInstaller
Oct 25 05:32:34 ip-172-31-45-194 jenkins[10699]: 2025-10-25 05:32:34.287+0000 [id=49] INFO hudson.util.Retrier$start: Performed the action check updates server successfully at the attempt #1
(venv) ubuntu@ip-172-31-45-194:~/terraform-aws-flask-express-deployment/apps/express_app$
```


Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log ([not sure where to find it?](#)) and this file on the server:

```
/var/lib/jenkins/secrets/initialAdminPassword
```

Please copy the password from either location and paste it below.

Administrator password

Continue

Getting Started



✗ Folders	✓ OWASP Markup Formatter	🔄 Build Timeout	🔄 Credentials Binding	** commons-lang3 v3.x Jenkins API ** Ionicons API
🔄 Timestamper	🔄 Workspace Cleanup	🔄 Ant	🔄 Gradle	Folders
🔄 Pipeline	🔄 GitHub Branch Source	🔄 Pipeline: GitHub Groovy Libraries	🔄 Pipeline Graph View	OWASP Markup Formatter
🔄 Git	🔄 SSH Build Agents	🔄 Matrix Authorization Strategy	○ LDAP	
🔄 Email Extension	○ Mailer	🔄 Dark Theme		
				** - required dependency

Getting Started

✗ Folders	✓ OWASP Markup Formatter	✓ Build Timeout	✓ Credentials Binding	** Resource Disposer Workspace Cleanup Ant ** OkHttp ** Apache HttpComponents Client 4.x API ** Pipeline: Job Gradle ** Pipeline: Milestone Step ** Durable Task ** Pipeline: Nodes and Processes ** Pipeline: Build Step ** Pipeline: SCM Step ** Pipeline: Groovy ** Pipeline: Groovy Libraries ** Joda Time API ** Pipeline: Model API ** Pipeline: Stage Step ** Pipeline: Declarative Extension Points API ** Jakarta Mail API ** Instance Identity Mailer ** Branch API ** Pipeline: Multibranch ** Pipeline: Stage Tags Metadata ** Pipeline: Input Step ** Pipeline: Basic Steps ** Pipeline: Declarative Pipeline ** Java JSON Web Token (JWT) ** GitHub API ** Mina SSHD API :: Common ** Mina SSHD API :: Core ** Gson API ** - required dependency
✗ Timestamper	✗ Workspace Cleanup	✗ Ant	✗ Gradle	
✗ Pipeline	🔄 GitHub Branch Source	🔄 Pipeline: GitHub Groovy Libraries	🔄 Pipeline Graph View	
🔄 Git	🔄 SSH Build Agents	🔄 Matrix Authorization Strategy	🔄 LDAP	
🔄 Email Extension	✓ Mailer	🔄 Dark Theme		

Getting Started

✓ Folders	✓ Timestamper	✓ Prism API	✓ JUnit	Gson API
✓ Matrix Project	✓ Workspace Cleanup	✓ Ant	✓ OkHttp	Git client
✓ Gradle	✓ Pipeline: Groovy Libraries	✓ Branch API	✓ Pipeline: Multibranch	Git
✗ Pipeline: Declarative	✗ Pipeline	✗ GitHub API	✗ Gson API	GitHub
✗ Git client	✗ Git	✗ GitHub	✗ GitHub Branch Source	GitHub Branch Source
✗ Pipeline: GitHub Groovy Libraries	✗ Pipeline Graph View	✗ Trilead API	✗ SSH Build Agents	Pipeline: GitHub Groovy Libraries
✗ Email Extension				** Metrics
				Pipeline Graph View
				Git
				** EDDSA API
				Trilead API
				SSH Build Agents
				** Matrix Authorization Strategy
				** LDAP
				** jsoup API
				Email Extension
				** Mailer
				** Theme Manager
				** Dark Theme
				Folders
				Timestamper
				Prism API
				JUnit
				Matrix Project
				Workspace Cleanup
				Ant
				OkHttp
				Gradle
				Pipeline: Groovy Libraries
				Branch API
				Pipeline: Multibranch
				** - required dependency

Create First Admin User

Username

Password

Confirm password

Full name

E-mail address


```
Get:54 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libalgorithm-diff-xs-perl amd64 0.04-8build3 [11.2 kB]
Get:55 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libalgorithm-merge-perl all 0.08-5 [11.4 kB]
Get:56 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libaom3 amd64 3.8.2-2ubuntu0.1 [1941 kB]
Get:57 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libfontconfig1 amd64 2.15.0-1.1ubuntu2 [139 kB]
Get:58 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libsharpv0 amd64 1.3.2-0.4build3 [15.8 kB]
Get:59 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libheif-plugin-aomdec amd64 1.17.6-1ubuntu4.1 [10.4 kB]
Get:60 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libde265-0 amd64 1.0.15-1build3 [166 kB]
Get:61 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libheif-plugin-libde265 amd64 1.17.6-1ubuntu4.1 [8176 B]
Get:62 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libheif1 amd64 1.17.6-1ubuntu4.1 [275 kB]
Get:63 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libjpeg-turbo8 amd64 2.1.5-2ubuntu2 [150 kB]
Get:64 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libjpeg8 amd64 8c-2ubuntu11 [2148 B]
Get:65 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libdeflate0 amd64 1.19-1build1.1 [43.9 kB]
Get:66 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libjbig0 amd64 2.1-6.1ubuntu2 [29.7 kB]
Get:67 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 liblerc4 amd64 4.0.0+ds-4ubuntu2 [179 kB]
Get:68 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libwebp7 amd64 1.3.2-0.4build3 [230 kB]
Get:69 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libtiff6 amd64 4.5.1+git230720-4ubuntu2.4 [199 kB]
Get:70 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libxpm4 amd64 1:3.5.17-1build2 [36.5 kB]
Get:71 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libgd3 amd64 2.3.3-9ubuntu5 [128 kB]
Get:72 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libc-devtools amd64 2.39-0ubuntu8.6 [29.3 kB]
Get:73 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libexpat1-dev amd64 2.6.1-2ubuntu0.3 [140 kB]
Get:74 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libfile-fcntllock-perl amd64 0.22-4ubuntu5 [30.7 kB]
Get:75 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libheif-plugin-aomenc amd64 1.17.6-1ubuntu4.1 [14.7 kB]
Get:76 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libjs-jquery all 3.6.1+dfsg+~3.5.14-1 [328 kB]
Get:77 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libjs-underscore all 1.13.4~dfsg+~1.11.4-3 [118 kB]
Get:78 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libjs-sphinxdoc all 7.2.6-6 [149 kB]
Get:79 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 zlib1g-dev amd64 1:1.3.dfsg-3.1ubuntu2.1 [894 kB]
Get:80 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libpython3.12-dev amd64 3.12.3-1ubuntu0.8 [5677 kB]
Get:81 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libpython3-dev amd64 3.12.3-0ubuntu2 [10.3 kB]
Get:82 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 manpages-dev all 6.7-2 [2013 kB]
Get:83 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 python3.12-dev amd64 3.12.3-1ubuntu0.8 [498 kB]
Get:84 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 python3-dev amd64 3.12.3-0ubuntu2 [26.7 kB]
Get:85 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 python3-wheel all 0.42.0-2 [53.1 kB]
Get:86 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 python3-pip all 24.0+dfsg-1ubuntu1.3 [1320 kB]
Fetched 87.3 MB in 1s (92.2 MB/s)
Extracting templates from packages: 100%
Preconfiguring packages ...
(Reading database ... 71718 files and directories currently installed.)
Preparing to unpack .../dpkg_1.22.6ubuntu6.5_amd64.deb ...
Unpacking dpkg (1.22.6ubuntu6.5) over (1.22.6ubuntu6.1) ...
Setting up dpkg (1.22.6ubuntu6.5) ...
(Reading database ... 71718 files and directories currently installed.)
Preparing to unpack .../libc6_2.39-0ubuntu8.6_amd64.deb ...
Unpacking libc6:amd64 (2.39-0ubuntu8.6) over (2.39-0ubuntu8.5) ...
Setting up libc6:amd64 (2.39-0ubuntu8.6) ...
(Reading database ... 71718 files and directories currently installed.)
Preparing to unpack .../libc-bin_2.39-0ubuntu8.6_amd64.deb ...
Unpacking libc-bin (2.39-0ubuntu8.6) over (2.39-0ubuntu8.5) ...
Setting up libc-bin (2.39-0ubuntu8.6) ...
(Reading database ... 71718 files and directories currently installed.)
Preparing to unpack .../00-libpython3.12t64_3.12.3-1ubuntu0.8_amd64.deb ...
Unpacking libpython3.12t64:amd64 (3.12.3-1ubuntu0.8) over (3.12.3-1ubuntu0.7) ...
Preparing to unpack .../01-python3.12_3.12.3-1ubuntu0.8_amd64.deb ...
Unpacking python3.12 (3.12.3-1ubuntu0.8) over (3.12.3-1ubuntu0.7) ...
```

```
Unpacking libgcc-13-dev:amd64 (13.3.0-6ubuntu2~24.04) ...
Selecting previously unselected package gcc-13-x86-64-linux-gnu.
Preparing to unpack .../37-gcc-13-x86-64-linux-gnu_13.3.0-6ubuntu2~24.04_amd64.deb ...
Unpacking gcc-13-x86-64-linux-gnu (13.3.0-6ubuntu2~24.04) ...
Selecting previously unselected package gcc-13.
Preparing to unpack .../38-gcc-13_13.3.0-6ubuntu2~24.04_amd64.deb ...
Unpacking gcc-13 (13.3.0-6ubuntu2~24.04) ...
Selecting previously unselected package gcc-x86-64-linux-gnu.
Preparing to unpack .../39-gcc-x86-64-linux-gnu_4%3a13.2.0-7ubuntu1_amd64.deb ...
Unpacking gcc-x86-64-linux-gnu (4:13.2.0-7ubuntu1) ...
Selecting previously unselected package gcc.
Preparing to unpack .../40-gcc_4%3a13.2.0-7ubuntu1_amd64.deb ...
Unpacking gcc (4:13.2.0-7ubuntu1) ...
Selecting previously unselected package libstdc++-13-dev:amd64.
Preparing to unpack .../41-libstdc++-13-dev_13.3.0-6ubuntu2~24.04_amd64.deb ...
Unpacking libstdc++-13-dev:amd64 (13.3.0-6ubuntu2~24.04) ...
Selecting previously unselected package g++-13-x86-64-linux-gnu.
Preparing to unpack .../42-g++-13-x86-64-linux-gnu_13.3.0-6ubuntu2~24.04_amd64.deb ...
Unpacking g++-13-x86-64-linux-gnu (13.3.0-6ubuntu2~24.04) ...
Selecting previously unselected package g++-13.
Preparing to unpack .../43-g++-13_13.3.0-6ubuntu2~24.04_amd64.deb ...
Unpacking g++-13 (13.3.0-6ubuntu2~24.04) ...
Selecting previously unselected package g++-x86-64-linux-gnu.
Preparing to unpack .../44-g++-x86-64-linux-gnu_4%3a13.2.0-7ubuntu1_amd64.deb ...
Unpacking g++-x86-64-linux-gnu (4:13.2.0-7ubuntu1) ...
Selecting previously unselected package g++.
Preparing to unpack .../45-g++_4%3a13.2.0-7ubuntu1_amd64.deb ...
Unpacking g++ (4:13.2.0-7ubuntu1) ...
Selecting previously unselected package make.
Preparing to unpack .../46-make_4.3-4.1build2_amd64.deb ...
Unpacking make (4.3-4.1build2) ...
Selecting previously unselected package libdpkg-perl.
Preparing to unpack .../47-libdpkg-perl_1.22.6ubuntu6.5_all.deb ...
Unpacking libdpkg-perl (1.22.6ubuntu6.5) ...
Selecting previously unselected package bzip2.
Preparing to unpack .../48-bzip2_1.0.8-5.1build0.1_amd64.deb ...
Unpacking bzip2 (1.0.8-5.1build0.1) ...
Selecting previously unselected package lto-disabled-list.
Preparing to unpack .../49-lto-disabled-list_47_all.deb ...
Unpacking lto-disabled-list (47) ...
Selecting previously unselected package dpkg-dev.
Preparing to unpack .../50-dpkg-dev_1.22.6ubuntu6.5_all.deb ...
Unpacking dpkg-dev (1.22.6ubuntu6.5) ...
Selecting previously unselected package build-essential.
Preparing to unpack .../51-build-essential_12.10ubuntu1_amd64.deb ...
Unpacking build-essential (12.10ubuntu1) ...
Selecting previously unselected package libfakeroot:amd64.
Preparing to unpack .../52-libfakeroot_1.33-1_amd64.deb ...
Unpacking libfakeroot:amd64 (1.33-1) ...
Selecting previously unselected package fakeroot.
Preparing to unpack .../53-fakeroot_1.33-1_amd64.deb ...
Unpacking fakeroot (1.33-1) ...
Selecting previously unselected package fonts-dejavu-mono.
```

```
Setting up libc6-dev:amd64 (2.39-0ubuntu8.6) ...
Setting up libstdc++-13-dev:amd64 (13.3.0-6ubuntu2~24.04) ...
Setting up binutils-x86-64-linux-gnu (2.42-4ubuntu2.5) ...
Setting up cpp-x86-64-linux-gnu (4:13.2.0-7ubuntu1) ...
Setting up cpp-13 (13.3.0-6ubuntu2~24.04) ...
Setting up gcc-13-x86-64-linux-gnu (13.3.0-6ubuntu2~24.04) ...
Setting up binutils (2.42-4ubuntu2.5) ...
Setting up dpkg-dev (1.22.6ubuntu6.5) ...
Setting up libexpat1-dev:amd64 (2.6.1-2ubuntu0.3) ...
Setting up gcc-13 (13.3.0-6ubuntu2~24.04) ...
Setting up zlib1g-dev:amd64 (1:1.3.dfsg-3.1ubuntu2.1) ...
Setting up cpp (4:13.2.0-7ubuntu1) ...
Setting up g++-13-x86-64-linux-gnu (13.3.0-6ubuntu2~24.04) ...
Setting up gcc-x86-64-linux-gnu (4:13.2.0-7ubuntu1) ...
Setting up libpython3.12-dev:amd64 (3.12.3-1ubuntu0.8) ...
Setting up gcc (4:13.2.0-7ubuntu1) ...
Setting up python3.12-dev (3.12.3-1ubuntu0.8) ...
Setting up g++-x86-64-linux-gnu (4:13.2.0-7ubuntu1) ...
Setting up g++-13 (13.3.0-6ubuntu2~24.04) ...
Setting up libpython3-dev:amd64 (3.12.3-0ubuntu2) ...
Setting up python3-dev (3.12.3-0ubuntu2) ...
Setting up g++ (4:13.2.0-7ubuntu1) ...
update-alternatives: using /usr/bin/g++ to provide /usr/bin/c++ (c++) in auto mode
Setting up build-essential (12.10ubuntu1) ...
Setting up libheif-plugin-aomdec:amd64 (1.17.6-1ubuntu4.1) ...
Setting up libheif1:amd64 (1.17.6-1ubuntu4.1) ...
Setting up libheif-plugin-libde265:amd64 (1.17.6-1ubuntu4.1) ...
Setting up libheif-plugin-aomenc:amd64 (1.17.6-1ubuntu4.1) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
Processing triggers for systemd (255.4-1ubuntu8.10) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for sgml-base (1.31) ...
Setting up libfontconfig1:amd64 (2.15.0-1.1ubuntu2) ...
Setting up libgd3:amd64 (2.3.3-9ubuntu5) ...
Setting up libc-devtools (2.39-0ubuntu8.6) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...
systemctl restart acpid.service chrony.service cron.service fwupd.service irqbalance.service multipathd.service packagekit.service polkit.service rsyslog.service s
sh.service systemd-journald.service systemd-networkd.service systemd-resolved.service systemd-udev.service udisks2.service

Service restarts being deferred:
systemctl restart ModemManager.service
/etc/needrestart/restart.d/dbus.service
systemctl restart getty@tty1.service
systemctl restart networkd-dispatcher.service
systemctl restart serial-getty@ttyS0.service
systemctl restart systemd-logind.service
```

```

ubuntu@ip-172-31-45-194:~/terraform-aws-flask-express-deployment/apps/flask_app$ sudo apt update -y
sudo apt install -y python3.12-venv
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 https://deb.nodesource.com/node_18.x nodistro InRelease
Hit:5 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
45 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  python3-pip-whl python3-setuptools-whl
The following NEW packages will be installed:
  python3-pip-whl python3-setuptools-whl python3.12-venv
0 upgraded, 3 newly installed, 0 to remove and 45 not upgraded.
Need to get 2429 kB of archives.
After this operation, 2777 kB of additional disk space will be used.
Get:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 python3-pip-whl all 24.0+dfsg-1ubuntu1.3 [1787 kB]
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 python3-setuptools-whl all 68.1.2-2ubuntu1.2 [716 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 python3.12-venv amd64 3.12.3-1ubuntu0.8 [5678 B]
Fetched 2429 kB in 0s (58.4 MB/s)
Selecting previously unselected package python3-pip-whl.
(Reading database ... 84479 files and directories currently installed.)
Preparing to unpack .../python3-pip-whl_24.0+dfsg-1ubuntu1.3_all.deb ...
Unpacking python3-pip-whl (24.0+dfsg-1ubuntu1.3) ...
Selecting previously unselected package python3-setuptools-whl.
Preparing to unpack .../python3-setuptools-whl_68.1.2-2ubuntu1.2_all.deb ...
Unpacking python3-setuptools-whl (68.1.2-2ubuntu1.2) ...
Selecting previously unselected package python3.12-venv.
Preparing to unpack .../python3.12-venv_3.12.3-1ubuntu0.8_amd64.deb ...
Unpacking python3.12-venv (3.12.3-1ubuntu0.8) ...
Setting up python3-setuptools-whl (68.1.2-2ubuntu1.2) ...
Setting up python3-pip-whl (24.0+dfsg-1ubuntu1.3) ...
Setting up python3.12-venv (3.12.3-1ubuntu0.8) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service
systemctl restart getty@tty1.service
systemctl restart networkd-dispatcher.service
systemctl restart serial-getty@ttyS0.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

ubuntu@ip-172-31-45-194:~/terraform-aws-flask-express-deployment/apps/flask_app$ cd ~/terraform-aws-flask-express-deployment/apps/flask_app
python3 -m venv venv
source venv/bin/activate
pip install -r requirements.txt
Collecting Flask==2.2.5 (from -r requirements.txt (line 1))
  Downloading Flask-2.2.5-py3-none-any.whl.metadata (3.9 kB)
Collecting Werkzeug>=2.2.2 (from Flask==2.2.5->-r requirements.txt (line 1))
  Downloading werkzeug-3.1.3-py3-none-any.whl.metadata (3.7 kB)
Collecting Jinja2>=3.0 (from Flask==2.2.5->-r requirements.txt (line 1))
  Downloading jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting itsdangerous>=2.0 (from Flask==2.2.5->-r requirements.txt (line 1))
  Downloading itsdangerous-2.2.0-py3-none-any.whl.metadata (1.9 kB)
Collecting click>=8.0 (from Flask==2.2.5->-r requirements.txt (line 1))
  Downloading click-8.3.0-py3-none-any.whl.metadata (2.6 kB)
Collecting MarkupSafe>=2.0 (from Jinja2>=3.0->Flask==2.2.5->-r requirements.txt (line 1))
  Downloading markupsafe-3.0.3-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (2.7 kB)
Downloading Flask-2.2.5-py3-none-any.whl (101 kB)
101.8/101.8 kB 6.2 MB/s eta 0:00:00
Downloading click-8.3.0-py3-none-any.whl (107 kB)
107.3/107.3 kB 9.8 MB/s eta 0:00:00
Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
Downloading Jinja2-3.1.6-py3-none-any.whl (134 kB)
134.9/134.9 kB 13.3 MB/s eta 0:00:00
Downloading werkzeug-3.1.3-py3-none-any.whl (224 kB)
224.5/224.5 kB 20.4 MB/s eta 0:00:00
Downloading markupsafe-3.0.3-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (22 kB)
Installing collected packages: MarkupSafe, itsdangerous, click, Werkzeug, Jinja2, Flask
Successfully installed Flask-2.2.5 Jinja2-3.1.6 MarkupSafe-3.0.3 Werkzeug-3.1.3 click-8.3.0 itsdangerous-2.2.0
(venv) ubuntu@ip-172-31-45-194:~/terraform-aws-flask-express-deployment/apps/flask_app$ python app.py
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.31.45.194:5000
Press CTRL+C to quit

```

```
^C(venv) ubuntu@ip-172-31-45-194:~/terraform-aws-flask-express-deployment/apps/flask_app$ pm2 start "venv/bin/python app.py" --name flask-app
pm2 save
pm2 startup
[PM2] Starting /usr/bin/bash in fork_mode (1 instance)
[PM2] Done.



| id | name      | namespace | version | mode | pid  | uptime | 0 | status | cpu | mem   | user   | watching |
|----|-----------|-----------|---------|------|------|--------|---|--------|-----|-------|--------|----------|
| 0  | flask-app | default   | N/A     | fork | 6998 | 0s     | 0 | online | 0%  | 9.7mb | ubuntu | disabled |



[PM2] Saving current process list...
[PM2] Successfully saved in /home/ubuntu/.pm2/dump.pm2
[PM2] Init System found: systemd
[PM2] To setup the Startup Script, copy/paste the following command:
sudo env PATH=$PATH:/usr/bin /usr/lib/node_modules/pm2/bin/pm2 startup systemd -u ubuntu --hp /home/ubuntu
(venv) ubuntu@ip-172-31-45-194:~/terraform-aws-flask-express-deployment/apps/flask_app$

(venv) ubuntu@ip-172-31-45-194:~/terraform-aws-flask-express-deployment/apps/flask_app$ pm2 list



| id | name      | namespace | version | mode | pid  | uptime | 0 | status | cpu | mem    | user   | watching |
|----|-----------|-----------|---------|------|------|--------|---|--------|-----|--------|--------|----------|
| 0  | flask-app | default   | N/A     | fork | 6998 | 21s    | 0 | online | 0%  | 30.2mb | ubuntu | disabled |



(venv) ubuntu@ip-172-31-45-194:~/terraform-aws-flask-express-deployment/apps/flask_app$
```

Not secure 15.206.91.151:5000

Pretty print

("message":"hello from Flask running in Docker on port 5000!")

```
(venv) ubuntu@ip-172-31-45-194:~/terraform-aws-flask-express-deployment/apps/flask_app$ cd ~/terraform-aws-flask-express-deployment/apps/express_app
npm install
node index.js

added 73 packages, and audited 74 packages in 2s

14 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
npm notice
npm notice New major version of npm available! 10.8.2 -> 11.6.2
npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.6.2
npm notice To update run: npm install -g npm@11.6.2
npm notice
Express app listening on port 3000
```

```
(venv) ubuntu@ip-172-31-45-194:~/terraform-aws-flask-express-deployment/apps/express_app$ pm2 start "node index.js" --name express-app
pm2 save
pm2 startup
[PM2] Starting /usr/bin/bash in fork_mode (1 instance)
[PM2] Done.



| id | name        | namespace | version | mode | pid  | uptime | 0 | status | cpu | mem    | user   | watching |
|----|-------------|-----------|---------|------|------|--------|---|--------|-----|--------|--------|----------|
| 1  | express-app | default   | N/A     | fork | 7059 | 0s     | 0 | online | 0%  | 20.0mb | ubuntu | disabled |
| 0  | flask-app   | default   | N/A     | fork | 6998 | 3m     | 0 | online | 0%  | 30.2mb | ubuntu | disabled |



[PM2] Saving current process list...
[PM2] Successfully saved in /home/ubuntu/.pm2/dump.pm2
[PM2] Init System found: systemd
[PM2] To setup the Startup Script, copy/paste the following command:
sudo env PATH=$PATH:/usr/bin /usr/lib/node_modules/pm2/bin/pm2 startup systemd -u ubuntu --hp /home/ubuntu
```

(venv) ubuntu@ip-172-31-45-194:~/terraform-aws-flask-express-deployment/apps/express_app\$ pm2 list

id	name	namespace	version	mode	pid	uptime	0	status	cpu	mem	user	watching
1	express-app	default	N/A	fork	7059	21s	0	online	0%	52.5mb	ubuntu	disabled
0	flask-app	default	N/A	fork	6998	3m	0	online	0%	30.2mb	ubuntu	disabled

< > ↺

🔒 Not secure 15.206.91.151:3000

🔗 🛡️ ⚠️

Hello from Express App on port 3000

```
(venv) ubuntu@ip-172-31-45-194:~/terraform-aws-flask-express-deployment/apps/express_app$ sudo ufw status
Status: inactive
(venv) ubuntu@ip-172-31-45-194:~/terraform-aws-flask-express-deployment/apps/express_app$ sudo systemctl status jenkins
● Jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
   Active: active (running) since Sat 2025-10-25 05:32:32 UTC; 28min ago
     Main PID: 10699 (java)
       Tasks: 38 (limit: 1000)
      Memory: 288.6M (peak: 321.0M)
         CPU: 22.652s
      CGroup: /system.slice/jenkins.service
              └─10699 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080

Oct 25 05:32:26 ip-172-31-45-194 jenkins[10699]: [LF]> This may also be found at: /var/lib/jenkins/secrets/initialAdminPassword
Oct 25 05:32:26 ip-172-31-45-194 jenkins[10699]: [LF]>
Oct 25 05:32:26 ip-172-31-45-194 jenkins[10699]: [LF]> *****
Oct 25 05:32:26 ip-172-31-45-194 jenkins[10699]: [LF]> *****
Oct 25 05:32:26 ip-172-31-45-194 jenkins[10699]: [LF]> *****
Oct 25 05:32:32 ip-172-31-45-194 jenkins[10699]: 2025-10-25 05:32:32.240+0000 [id=30] INFO jenkins.InitReactorRunner$1onAttained: Completed initialization
Oct 25 05:32:32 ip-172-31-45-194 jenkins[10699]: 2025-10-25 05:32:32.269+0000 [id=23] INFO hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
Oct 25 05:32:32 ip-172-31-45-194 systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
Oct 25 05:32:34 ip-172-31-45-194 jenkins[10699]: 2025-10-25 05:32:34.286+0000 [id=40] INFO h.m.DownloadService$Downloadable#load: Obtained the updated data file for hudson.tasks.Maven.MavenInstaller
Oct 25 05:32:34 ip-172-31-45-194 jenkins[10699]: 2025-10-25 05:32:34.287+0000 [id=49] INFO hudson.util.Retrier#start: Performed the action check updates server successfully at the attempt #1
(venv) ubuntu@ip-172-31-45-194:~/terraform-aws-flask-express-deployment/apps/express_app$
```


Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log ([not sure where to find it?](#)) and this file on the server:

```
/var/lib/jenkins/secrets/initialAdminPassword
```

Please copy the password from either location and paste it below.

Administrator password

Continue

Getting Started



✗ Folders	✓ OWASP Markup Formatter	🔄 Build Timeout	🔄 Credentials Binding	** commons-lang3 v3.x Jenkins API ** Ionicons API
🔄 Timestamper	🔄 Workspace Cleanup	🔄 Ant	🔄 Gradle	Folders
🔄 Pipeline	🔄 GitHub Branch Source	🔄 Pipeline: GitHub Groovy Libraries	🔄 Pipeline Graph View	OWASP Markup Formatter
🔄 Git	🔄 SSH Build Agents	🔄 Matrix Authorization Strategy	○ LDAP	
🔄 Email Extension	○ Mailer	🔄 Dark Theme		
				** - required dependency

Getting Started

✗ Folders	✓ OWASP Markup Formatter	✓ Build Timeout	✓ Credentials Binding	** Resource Disposer Workspace Cleanup Ant ** OkHttp ** Apache HttpComponents Client 4.x API ** Pipeline: Job Gradle ** Pipeline: Milestone Step ** Durable Task ** Pipeline: Nodes and Processes ** Pipeline: Build Step ** Pipeline: SCM Step ** Pipeline: Groovy ** Pipeline: Groovy Libraries ** Joda Time API ** Pipeline: Model API ** Pipeline: Stage Step ** Pipeline: Declarative Extension Points API ** Jakarta Mail API ** Instance Identity Mailer ** Branch API ** Pipeline: Multibranch ** Pipeline: Stage Tags Metadata ** Pipeline: Input Step ** Pipeline: Basic Steps ** Pipeline: Declarative Pipeline ** Java JSON Web Token (JWT) ** GitHub API ** Mina SSHD API :: Common ** Mina SSHD API :: Core ** Gson API ** - required dependency
✗ Timestampers	✗ Workspace Cleanup	✗ Ant	✗ Gradle	
✗ Pipeline	🔄 GitHub Branch Source	🔄 Pipeline: GitHub Groovy Libraries	🔄 Pipeline Graph View	
🔄 Git	🔄 SSH Build Agents	🔄 Matrix Authorization Strategy	🔄 LDAP	
🔄 Email Extension	✓ Mailer	🔄 Dark Theme		

Getting Started

✓ Folders	✓ Timestamper	✓ Prism API	✓ JUnit	Gson API
✓ Matrix Project	✓ Workspace Cleanup	✓ Ant	✓ OkHttp	Git client
✓ Gradle	✓ Pipeline: Groovy Libraries	✓ Branch API	✓ Pipeline: Multibranch	Git
✗ Pipeline: Declarative	✗ Pipeline	✗ GitHub API	✗ Gson API	GitHub
✗ Git client	✗ Git	✗ GitHub	✗ GitHub Branch Source	GitHub Branch Source
✗ Pipeline: GitHub Groovy Libraries	✗ Pipeline Graph View	✗ Trilead API	✗ SSH Build Agents	Pipeline: GitHub Groovy Libraries
✗ Email Extension				** Metrics
				Pipeline Graph View
				Git
				** EDDSA API
				Trilead API
				SSH Build Agents
				** Matrix Authorization Strategy
				** LDAP
				** jsoup API
				Email Extension
				** Mailer
				** Theme Manager
				** Dark Theme
				Folders
				Timestamper
				Prism API
				JUnit
				Matrix Project
				Workspace Cleanup
				Ant
				OkHttp
				Gradle
				Pipeline: Groovy Libraries
				Branch API
				Pipeline: Multibranch
				** - required dependency

Create First Admin User

Username

Password

Confirm password

Full name

E-mail address

Instance Configuration

Jenkins URL:

The Jenkins URL is used to provide the root URL for absolute links to various Jenkins resources. That means this value is required for proper operation of many Jenkins features including email notifications, PR status updates, and the BUILD_URL environment variable provided to build steps.

The proposed default value shown is **not saved yet** and is generated from the current request, if possible. The best practice is to set this value to the URL that users are expected to use. This will avoid confusion when sharing or viewing links.

Plugins

- Updates
- Available plugins
- Installed plugins
- Advanced settings
- Download progress

Download progress

Preparation	- Checking internet connectivity - Checking update center connectivity - Success
Pyenv Pipeline	Success
ShiningPanda	Pending
Loading plugin extensions	Pending
Restarting Jenkins	Pending

- [Go back to the top page](#)
(you can start using the installed plugins right away)
- ☒ Restart Jenkins when installation is complete and no jobs are running

NodeJS

Name

Node18

☒ Install automatically ?

Install from nodejs.org

Version

NodeJS 25.0.0

For the underlying architecture, if available, force the installation of the 32bit package. Otherwise the build will fail

☐ Force 32bit architecture

Global npm packages to install

Specify list of packages to install globally -- see npm install -g. Note that you can fix the packages version by using the syntax 'packageName@version'

Global npm packages refresh hours

Duration, in hours, before 2 npm cache update. Note that 0 will always update npm cache

72

+ Add Installer

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS

```
Setting up node-json5 (2.2.3+dfsg-1build1) ...
Setting up node-ms (2.1.3+~cs0.7.31-3) ...
Setting up node-floraf (3.0.2-2) ...
Setting up node-string-width (4.2.3+~cs13.2.3-1) ...
Setting up node-regenerate (1.4.2-3) ...
Setting up node-samever (7.5.4+~7.5.0-2) ...
Setting up node-pixcolor (1.0.0-4) ...
Setting up node-es-abstract (1.20.4+~cs26.27.47-1) ...
Setting up node-builtins (5.0.1-1) ...
Setting up node-is-primitive (3.0.1-3) ...
Setting up node-log-driver (1.2.7+git20180219bba1761737-7) ...
Setting up node-handledby (3.4.7+~4.1.0-1) ...
Setting up node-object-assign (4.1.1-7) ...
Setting up node-is-glob (4.0.3-2) ...
Setting up node-v8-compile-cache (2.4.0-2) ...
Setting up node-flat-cache (3.0.4-6.0.1+~3.0.4+~2.0.0+~1.0.0+~2.0.1-1) ...
Setting up node-is-number (7.0.0-4) ...
Setting up node-for-in (1.0.2-4) ...
Setting up node-webassemblyjs (1.11.4+dfsg+~cs10.11.17-2) ...
Setting up node-strip-json-comments (4.0.0-4) ...
Setting up libnode-dev (18.19.1+dfsg-6ubuntu5) ...
Setting up node-is-extendable (1.0.1-3) ...
Setting up node-columnify (1.6.0+~4.5.1-1) ...
Setting up node-doctrine (3.0.0-4) ...
Setting up node-copy-concurrently (1.0.5-9) ...
Setting up node-set-value (4.1.0+~4.0.1-2) ...
Setting up node-ajv (8.12.0+~8+~1.1-4) ...
Setting up node-jscs (3.0.2+~3.0.1-1) ...
Setting up node-regparser (0.9.1+ds-2) ...
Setting up node-errno (1.0.0-3) ...
Setting up node-decamelize (4.0.0-2) ...
Setting up node-move-concurrently (1.0.1-5) ...
Setting up node-normalize-path (3.0.0+~3.0.0-1) ...
Setting up node-functional-red-black-tree (1.0.1+20181105-7) ...
Setting up node-debug (4.3.4+~cs4.1.7-1) ...
Setting up node-agent-base (7.1.0+~8-2023071921-5) ...
Setting up node-browserslist (4.22.1+~cs6.1.28-1) ...
Setting up node-yargs-parser (21.1.1+~21.0.0-4) ...
Setting up node-coveralls (3.1.1-2build1) ...
Setting up node-make-dir (3.1.0-3) ...
Setting up node-validate-npm-package-name (5.0.0+~4.0.0-1) ...
Setting up node-esrecurse (4.3.0-2) ...
Setting up node-tap-mocha-reporter (5.0.3+~2.0.2-2) ...
Setting up node-file-entry-cache (6.0.1+~3.0.4+~2.0.0+~1.0.0+~2.0.1-1) ...
Setting up node-wrap-ansi (8.0.1+~8.0.1-3) ...
Setting up node-normalize-package-data (4.0.1+~2.4.1-1) ...
Setting up node-for-own (1.0.0-3) ...
Setting up node-eslint-scope (7.1.1+~7.4.1) ...
Setting up node-escodegen (2.1.0+dfsg+~0.8-1) ...
Setting up node-widest-line (3.1.0-2) ...
Setting up node-mixin-deep (2.0.1-3) ...
Setting up node-memory-fs (0.5.0+~0.3.3-2) ...
Setting up node-babel-helper-define-polyfill-provider (0.3.3+~0-20220913+ds1-1) ...
Setting up node-cli-table (0.3.11+~cs0.13.4-3) ...
```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS

```
6.0.13
ubuntu@ip-172-31-47-117:~$ git --version
git version 2.40.0
ubuntu@ip-172-31-47-117:~$ sudo apt install -y git
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
git is already the newest version (1:2.43.0-1ubuntu7.3).
0 upgraded, 0 newly installed, 0 to remove and 23 not upgraded.
ubuntu@ip-172-31-47-117:~$ cd -
git clone https://github.com/0079567603568sahal/terraform-aws-flask-express-deployment.git
cd terraform-aws-flask-express-deployment
Cloning into 'terraform-aws-flask-express-deployment'...
remote: Enumerating objects: 118, done.
remote: Counting objects: 100% (118/118), done.
remote: Compressing objects: 100% (109/109), done.
remote: Total 118 (delta 10), reused 46 (delta 7), pack-reused 0 (from 0)
Receiving objects: 100% (118/118), 6.24 MiB | 17.41 MiB/s, done.
Resolving deltas: 100% (10/10), done.
ubuntu@ip-172-31-47-117:~/terraform-aws-flask-express-deployment$ ls
README.md      Terraform AWS ECS Fargate Flask + Express Deployment.pdf  es          part2_separate_ec2  scripts          yes
README.md.txt  apps                                                       part1_single_ec2  part3_ecr_ecs      'terraform screen shot'
ubuntu@ip-172-31-47-117:~/terraform-aws-flask-express-deployment$ cd flask-backend
pip install -r requirements.txt
nohup python3 app.py &
-bash: cd: flask-backend: No such file or directory
error: externally-managed-environment

x This environment is externally managed
-> To install Python packages system-wide, try apt install
python3-xyz, where xyz is the package you are trying to
install.

If you wish to install a non-Debian-packaged Python package,
create a virtual environment using python3 -m venv path/to/venv.
Then use path/to/venv/bin/python and path/to/venv/bin/pip. Make
sure you have python3-full installed.

If you wish to install a non-Debian packaged Python application,
it may be easiest to use pipx install xyz, which will manage a
virtual environment for you. Make sure you have pipx installed.

See /usr/share/doc/python3.12/README.venv for more information.

note: If you believe this is a mistake, please contact your Python installation or OS distribution provider. You can override this, at the risk of breaking your Python installation or OS, by passing --break-system-packages.
hint: See PEP 668 for the detailed specification.
[1] 68340
ubuntu@ip-172-31-47-117:~/terraform-aws-flask-express-deployment$ nohup: ignoring input and appending output to 'nohup.out'
ls
ls
ls
README.md      Terraform AWS ECS Fargate Flask + Express Deployment.pdf  es          part1_single_ec2  part3_ecr_ecs      'terraform screen shot'
README.md.txt  apps                                                       nohup.out   part2_separate_ec2  scripts          yes
[1]+  Exit 2              nohup python3 app.py
ubuntu@ip-172-31-47-117:~/terraform-aws-flask-express-deployment$ cd apps/flask_app
ubuntu@ip-172-31-47-117:~/terraform-aws-flask-express-deployment/apps/flask_app$ ls
Dockerfile app.py package-lock.json package.json requirements.txt
```



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
ubuntu@ip-172-31-47-117:~/terraform-aws-flask-express-deployment/apps/flask_app$ ls
Dockerfile app.py package-lock.json package.json requirements.txt
ubuntu@ip-172-31-47-117:~/terraform-aws-flask-express-deployment/apps/flask_app$ python3 -m venv venv
source venv/bin/activate
The virtual environment was not created successfully because ensurepip is not
available. On Debian/Ubuntu systems, you need to install the python3-venv
package using the following command.

    apt install python3.12-venv

You may need to use sudo with that command. After installing the python3-venv
package, recreate your virtual environment.

Failing command: /home/ubuntu/terraform-aws-flask-express-deployment/apps/flask_app/venv/bin/python3

-bash: venv/bin/activate: No such file or directory
ubuntu@ip-172-31-47-117:~/terraform-aws-flask-express-deployment/apps/flask_app$ pip install --break
Defaulting to user installation because normal site-packages is not writeable
ERROR: You must give at least one requirement to install (see "pip help install")
ubuntu@ip-172-31-47-117:~/terraform-aws-flask-express-deployment/apps/flask_app$ sudo apt install -y python3.12-venv
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
python3-pip-whl python3-setuptools-whl
The following NEW packages will be installed:
python3-pip-whl python3-setuptools-whl python3.12-venv
0 upgraded, 3 newly installed, 0 to remove and 23 not upgraded.
Need to get 2429 kB of archives.
After this operation, 2777 kB of additional disk space will be used.
Get:1 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 python3-pip-whl all 24.0+dfsg-1ubuntu1.3 [1707 kB]
Get:2 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 python3-setuptools-whl all 68.1.2-2ubuntu1.2 [716 kB]
Get:3 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 python3.12-venv amd64 3.12.3-1ubuntu0.8 [5678 B]
Fetched 2429 kB in 1s (24.6 MB/s)
Selecting previously unselected package python3-pip-whl.
(Reading database ... 158602 files and directories currently installed.)
Preparing to unpack .../python3-pip-whl_24.0+dfsg-1ubuntu1.3_all.deb ...
Unpacking python3-pip-whl (24.0+dfsg-1ubuntu1.3) ...
Selecting previously unselected package python3-setuptools-whl.
Preparing to unpack .../python3-setuptools-whl_68.1.2-2ubuntu1.2_all.deb ...
Unpacking python3-setuptools-whl (68.1.2-2ubuntu1.2) ...
Selecting previously unselected package python3.12-venv.
Preparing to unpack .../python3.12-venv_3.12.3-1ubuntu0.8_amd64.deb ...
Unpacking python3.12-venv (3.12.3-1ubuntu0.8) ...
Setting up python3-setuptools-whl (68.1.2-2ubuntu1.2) ...
Setting up python3-pip-whl (24.0+dfsg-1ubuntu1.3) ...
Setting up python3.12-venv (3.12.3-1ubuntu0.8) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Pending kernel upgrade!
Running kernel version:
6.14.0-1011-aws
Diagnostics:
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Restarting services...

Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service
systemctl restart getty@tty1.service
systemctl restart networkd-dispatcher.service
systemctl restart serial-getty@tty0.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-47-117:~/terraform-aws-flask-express-deployment/apps/flask_app$ python3 -m venv venv
source venv/bin/activate
(venv) ubuntu@ip-172-31-47-117:~/terraform-aws-flask-express-deployment/apps/flask_app$ pip install -r requirements.txt
Collecting Flask==2.2.5 (from -r requirements.txt (line 1))
  Downloading Flask-2.2.5-py3-none-any.whl.metadata (3.9 kB)
Collecting Werkzeug==2.2.2 (from Flask==2.2.5->-r requirements.txt (line 1))
  Downloading Werkzeug-2.2.2-py3-none-any.whl.metadata (3.7 kB)
Collecting Jinja2==3.1.6 (from Flask==2.2.5->-r requirements.txt (line 1))
  Downloading Jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting itsdangerous==2.0 (from Flask==2.2.5->-r requirements.txt (line 1))
  Downloading itsdangerous-2.0-py3-none-any.whl.metadata (1.9 kB)
Collecting click==8.0 (from Flask==2.2.5->-r requirements.txt (line 1))
  Downloading click-8.3.0-py3-none-any.whl.metadata (2.6 kB)
Collecting MarkupSafe==2.0 (from Jinja2==3.1.6->-r requirements.txt (line 1))
  Downloading MarkupSafe-2.0.3-cp312-cp312-manylinux2014_x86_64.manylinux2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (2.7 kB)
Downloading Flask-2.2.5-py3-none-any.whl (101 kB)
101.5/101.8 kB 6.2 MB/s eta 0:00:00
Downloading click-8.3.0-py3-none-any.whl (107 kB)
107.3/107.3 kB 9.4 MB/s eta 0:00:00
Downloading itsdangerous-2.0-py3-none-any.whl (16 kB)
Downloading Jinja2-3.1.6-py3-none-any.whl (134 kB)
134.9/134.9 kB 12.3 MB/s eta 0:00:00
Downloading Werkzeug-2.2.2-py3-none-any.whl (224 kB)
224.5/224.5 kB 10.3 MB/s eta 0:00:00
Downloading markupsafe-2.0.3-cp312-cp312-manylinux2014_x86_64.manylinux2_17_x86_64.manylinux_2_28_x86_64.whl (22 kB)
Installing collected packages: MarkupSafe, itsdangerous, click, Werkzeug, Jinja2, Flask
Successfully installed Flask-2.2.5 Jinja2-3.1.6 MarkupSafe-2.0.3 Werkzeug-3.1.3 click-8.3.0 itsdangerous-2.0
(venv) ubuntu@ip-172-31-47-117:~/terraform-aws-flask-express-deployment/apps/flask_app$ pip install --break-system-packages -r requirements.txt
Requirement already satisfied: Flask==2.2.5 in ./venv/lib/python3.12/site-packages (from -r requirements.txt (line 1)) (2.2.5)
Requirement already satisfied: Werkzeug==2.2.2 in ./venv/lib/python3.12/site-packages (from Flask==2.2.5->-r requirements.txt (line 1)) (3.1.3)
Requirement already satisfied: Jinja2==3.0 in ./venv/lib/python3.12/site-packages (from Flask==2.2.5->-r requirements.txt (line 1)) (3.1.6)
Requirement already satisfied: itsdangerous==2.0 in ./venv/lib/python3.12/site-packages (from Flask==2.2.5->-r requirements.txt (line 1)) (2.2.0)
Requirement already satisfied: click==8.0 in ./venv/lib/python3.12/site-packages (from Flask==2.2.5->-r requirements.txt (line 1)) (8.3.0)
Requirement already satisfied: MarkupSafe==2.0 in ./venv/lib/python3.12/site-packages (from Jinja2==3.0->-r requirements.txt (line 1)) (3.0.3)
(venv) ubuntu@ip-172-31-47-117:~/terraform-aws-flask-express-deployment/apps/flask_app$ nohup python3 app.py &
[1] 68624
(venv) ubuntu@ip-172-31-47-117:~/terraform-aws-flask-express-deployment/apps/flask_app$ nohup: ignoring input and appending output to 'nohup.out'
ps aux | grep app.py
ubuntu 68624 2.2 3.2 40528 30852 pts/0 S 11:43 0:00 python3 app.py
ubuntu 68626 0.0 0.2 7080 2080 pts/0 S+ 11:43 0:00 grep --color=auto app.py
```



OAuth Apps

Personal access tokens

Fine-grained tokens

Tokens (classic)

New personal access token (classic)

Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to [authenticate to the API over Basic Authentication](#).

Note

What's this token for?

Expiration

30 days (Nov 26, 2025)

The token will expire on the selected date.

Select scopes

Scopes define the access for personal tokens. [Read more about OAuth scopes](#).

☐ repo	Full control of private repositories
☐ repo:status	Access commit status
☐ repo_deployment	Access deployment status
☐ public_repo	Access public repositories
☐ repo:invite	Access repository invitations
☐ security_events	Read and write security events
☒ workflow	Update GitHub Action workflows
☐ write:packages	Upload packages to GitHub Package Registry
☐ read:packages	Download packages from GitHub Package Registry
☐ delete:packages	Delete packages from GitHub Package Registry
☐ admin:org	Full control of orgs and teams, read and write org projects
☐ write:org	Read and write org and team membership, read and write org projects
☐ read:org	Read org and team membership, read org projects
☐ manage_runners:org	Manage org runners and runner groups
☐ admin:public_key	Full control of user public keys
☐ write:public_key	Write user public keys
☐ read:public_key	Read user public keys
☒ admin:repo_hook	Full control of repository hooks
☐ write:repo_hook	Write repository hooks

Jenkins / Flask-Backend-CI-CD / #15

Status

Changes

Console Output

Edit Build Information

Timings

Git Build Data

Pipeline Overview

Thread Dump

Pause/resume

Replay

Pipeline Steps

Workspaces

Previous Build

#15 (27 Oct 2025, 08:08:36)

Started by user MUHAMMED SAHAL SR

This run spent 10 ms waiting in the queue.

git

Revision: 247f24a2c5cc7c1f7fa737bb90e150c1525a20bb

Repository: <https://github.com/0079567603568sahal/flask-express-cicd-deployment.git>

• refs/remotes/origin/main

No changes.

Progress:

Add description

Keep this build forever

Started 1 min 25 sec ago

Build has been executing for 1 min 25 sec

REST API Jenkins 2.528.1

Jenkins installation fix x 3.25.76.39-5000 x Flask-Backend-CI-CD #25 flask-express-cicd-deploy 007956760356bsahal/Flask flask-express-cicd-deploy Instance details | EC2 | ap Jenkins CI/CD Pipeline - G DEVOPS Lecture Page

< > ⌂ ⚠ Not secure 3.25.76.39-5000

Pretty print

```
{ "message": "Hello from flask backend!" }
```

 **Jenkins** / Express-Frontend-CI-CD / #1

Status

Changes

Console Output

Edit Build Information

Delete build #1

Timings

Git Build Data

Pipeline Overview

Restart from Stage

Replay

Pipeline Steps

Workspaces

✓ #1 (27 Oct 2025, 08:59:07)

Started by user MUHAMMED SAHAL SR

This run spent:

- 18 ms waiting;
- 17 sec build duration;
- 17 sec total from scheduled to completion.

git

Revision: 09ac046674b1194b9a66d91431082ff12c833ce8
Repository: <https://github.com/007956760356bsahal/flask-express-cicd-deployment.git>

- refs/remotes/origin/main

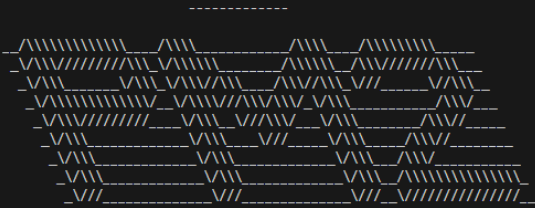
</> No changes.

Add description

Keep this build forever

Started 20 sec ago
Took 17 sec

Last login: Mon Oct 27 06:56:59 2025 from 111.92.27.39
ubuntu@ip-172-31-40-34:~\$ pm2 list



Runtime Edition

PM2 is a Production Process Manager for Node.js applications
with a built-in Load Balancer.

Start and Daemonize any application:
\$ pm2 start app.js

Load Balance 4 instances of api.js:
\$ pm2 start api.js -i 4

Monitor in production:
\$ pm2 monitor

Make pm2 auto-boot at server restart:
\$ pm2 startup

To go further checkout:
<http://pm2.io/>

[PM2] Spawning PM2 daemon with pm2_home=/home/ubuntu/.pm2
[PM2] PM2 Successfully daemonized

id	name	namespace	version	mode	pid	uptime	↓	status	cpu	mem	user	watching
----	------	-----------	---------	------	-----	--------	---	--------	-----	-----	------	----------

ubuntu@ip-172-31-40-34:~\$ pm2 list

id	name	namespace	version	mode	pid	uptime	↓	status	cpu	mem	user	watching
----	------	-----------	---------	------	-----	--------	---	--------	-----	-----	------	----------

ubuntu@ip-172-31-40-34:~\$

```
ubuntu@ip-172-31-40-34:/var/lib/jenkins/workspace/Flask-Backend-CI-CD/terraform-aws-flask-express-deployment/Flask-Backend$ python3 -m pip install --break-system-packages -r requirements.txt
Defaulting to user installation because normal site-packages is not writeable
Collecting Flask==2.3.3 (from -r requirements.txt (line 1))
  Downloading flask-2.3.3-py3-none-any.whl.metadata (3.6 kB)
Collecting gunicorn==21.2.0 (from -r requirements.txt (line 2))
  Downloading gunicorn-21.2.0-py3-none-any.whl.metadata (4.1 kB)
Collecting Werkzeug==2.3.7 (from Flask==2.3.3->-r requirements.txt (line 1))
  Using cached werkzeug-3.1.3-py3-none-any.whl.metadata (3.7 kB)
Requirement already satisfied: Jinja2>=3.1.2 in /usr/lib/python3/dist-packages (from Flask==2.3.3->-r requirements.txt (line 1)) (3.1.2)
Collecting itsdangerous>=2.1.2 (from Flask==2.3.3->-r requirements.txt (line 1))
Requirement already satisfied: click>=8.1.3 in /usr/lib/python3/dist-packages (from Flask==2.3.3->-r requirements.txt (line 1)) (8.1.6)
Requirement already satisfied: blinker>=1.6.2 in /usr/lib/python3/dist-packages (from Flask==2.3.3->-r requirements.txt (line 1)) (1.7.0)
Requirement already satisfied: packaging in /usr/lib/python3/dist-packages (from gunicorn==21.2.0->-r requirements.txt (line 2)) (24.0)
Requirement already satisfied: MarkupSafe>=2.1.1 in /usr/lib/python3/dist-packages (from Werkzeug==2.3.7->Flask==2.3.3->-r requirements.txt (line 1)) (2.1.5)
Downloading flask-2.3.3-py3-none-any.whl (96 kB)
   96.1/96.1 kB 4.2 MB/s eta 0:00:00
Downloading gunicorn-21.2.0-py3-none-any.whl (88 kB)
   88.0/88.2 kB 7.5 MB/s eta 0:00:00
Using cached itsdangerous-2.2.0-py3-none-any.whl (16 kB)
Using cached werkzeug-3.1.3-py3-none-any.whl (224 kB)
Installing collected packages: Werkzeug, itsdangerous, gunicorn, Flask
  WARNING: The script gunicorn is installed in '/home/ubuntu/.local/bin' which is not on PATH.
    Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
  WARNING: The script flask is installed in '/home/ubuntu/.local/bin' which is not on PATH.
    Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed Flask-2.3.3 Werkzeug-3.1.3 gunicorn-21.2.0 itsdangerous-2.2.0
ubuntu@ip-172-31-40-34:/var/lib/jenkins/workspace/Flask-Backend-CI-CD/terraform-aws-flask-express-deployment/Flask-Backend$ python3 -m flask --version
Python 3.12.3
Flask 2.3.3
Werkzeug 3.1.3
ubuntu@ip-172-31-40-34:/var/lib/jenkins/workspace/Flask-Backend-CI-CD/terraform-aws-flask-express-deployment/Flask-Backend$ python3 app.py
 * Serving Flask app 'app'
 * Debug mode: off
Address already in use
Port 5000 is in use by another program. Either identify and stop that program, or start the server with a different port.
ubuntu@ip-172-31-40-34:/var/lib/jenkins/workspace/Flask-Backend-CI-CD/terraform-aws-flask-express-deployment/Flask-Backend$ pm2 start app.py --name flask-backend --interpreter python3
pm2 save
[PM2] Starting /var/lib/jenkins/workspace/Flask-Backend-CI-CD/terraform-aws-flask-express-deployment/Flask-Backend/app.py in fork_mode (1 instance)
[PM2] Done.
```

id	name	namespace	version	mode	pid	uptime	↓	status	cpu	mem	user	watching
0	flask-backend	default	N/A	fork	2376	0s	0	online	0%	3.0mb	ubuntu	disabled

[PM2] Saving current process list...

[PM2] Successfully saved in /home/ubuntu/.pm2/dmmpm2

```
ubuntu@ip-172-31-40-34:/var/lib/jenkins/workspace/Flask-Backend-CI-CD/terraform-aws-flask-express-deployment/Flask-Backend$ sudo ss -ltn -i :5000
COMMAND  PID USER  FD  TYPE DEVICE SIZE/OFF NODE NAME
docker-pr 1375 root   7u  IPv4  8583   0t0  TCP *:5000 (LISTEN)
docker-pr 1381 root   7u  IPv6  8584   0t0  TCP *:5000 (LISTEN)
ubuntu@ip-172-31-40-34:/var/lib/jenkins/workspace/Flask-Backend-CI-CD/terraform-aws-flask-express-deployment/Flask-Backend$ sudo docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS                               NAMES
def24d3ae349   aws-flask-express-deploy_backend   "gunicorn -w 3 -b 0.0..." 21 minutes ago Up 21 minutes 0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp   aws-flask-express-deploy_backend_1
b8c6ee707ffe   aws-flask-express-deploy_frontend  "docker-entrypoint.sh"    21 minutes ago Up 21 minutes 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp   aws-flask-express-deploy_frontend_1
```

```

ubuntu@ip-172-31-48-34:/var/lib/jenkins/workspace/Flask-Backend-CI-CD/terraform-aws-flask-express-deployment/Flask-Backend$ sudo docker logs def24d3ae349 --tail 20
sudo docker logs b8c6ee707ffe --tail 20
[2025-10-27 08:52:38 +0000] [1] [INFO] Starting gunicorn 23.0.0
[2025-10-27 08:52:38 +0000] [1] [INFO] Listening at: http://0.0.0.0:5000 (1)
[2025-10-27 08:52:38 +0000] [1] [INFO] Using worker: sync
[2025-10-27 08:52:38 +0000] [7] [INFO] Booting worker with pid: 7
[2025-10-27 08:52:38 +0000] [8] [INFO] Booting worker with pid: 8
[2025-10-27 08:52:38 +0000] [9] [INFO] Booting worker with pid: 9
Express server running on port 5000

ubuntu@ip-172-31-48-34:/var/lib/jenkins/workspace/Flask-Backend-CI-CD/terraform-aws-flask-express-deployment/Flask-Backend$ sudo docker ps --format '{{.ID}}\t{{.Names}}\t{{.Status}}'
CONTAINER ID        NAMES                                     STATUS
def24d3ae349        aws-flask-express-deploy_backend_1      Up 23 minutes
b8c6ee707ffe        aws-flask-express-deploy_frontend_1     Up 23 minutes

ubuntu@ip-172-31-48-34:/var/lib/jenkins/workspace/Flask-Backend-CI-CD/terraform-aws-flask-express-deployment/Flask-Backend$ sudo docker update --restart always def24d3ae349
sudo docker update --restart always b8c6ee707ffe
def24d3ae349
b8c6ee707ffe

```

```

❖ Mohds@Mohdsahal MINGW64 ~/Downloads (main)
○ $ ssh -i "/c/Users/Mohds/Downloads/mohdsahal924 (1).pem" ubuntu@15.206.91.151
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1011-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Oct 25 05:01:46 UTC 2025

System load:  0.0           Temperature:   -273.1 C
Usage of /:   31.2% of 6.71GB Processes:      113
Memory usage: 25%          Users logged in: 0
Swap usage:   0%           IPv4 address for ens5: 172.31.45.194

 * Ubuntu Pro delivers the most comprehensive open source security and
   compliance features.

   https://ubuntu.com/aws/pro

Expanded Security Maintenance for Applications is not enabled.

61 updates can be applied immediately.
38 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-45-194:~$ sudo apt update -y
sudo apt install -y python3 python3-pip git curl
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
55 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done

```